Tate Frobenius on the $K(n)$-local sphere

October 27, 2026

Abstract

The Tate Frobenius of the $K(n)$-local sphere compares the chromatic overlap $L_{n-1}L_{K(n)}\mathbb{S}$ with the Tate construction $(L_{K(n)}\mathbb{S})^{t\Sigma_p}$ for the trivial action. We show that this comparison is a rational equivalence, that it has a retraction after $K(t)$-localization for every $1 \leq t < n$, and that it is an equivalence for $n \leq 2$. The retraction comes from the trace of a finite étale cover of degree prime to p, normalized by its value on the unit after descent. At $p = 2$ and $n = 3$ the comparison is not an equivalence, even after $K(1)$-localization and reduction modulo 2. The obstruction is an Artin–Schreier class attached to lifts modulo 4 of lines in the étale Tate module, and it is detected by a transfer.

1 Introduction

Fix a prime p and an integer $n \geq 1$, and write L_m for Bousfield localization at $K(0) \vee \cdots \vee K(m)$, where $K(0) = H\mathbb{Q}$. The $K(n)$-local sphere $L_{K(n)}\mathbb{S}$ is an $\mathbb{E}_\infty$-ring, so it has a Tate Frobenius

$$L_{K(n)}\mathbb{S} \longrightarrow \left((L_{K(n)}\mathbb{S})^{tC_p}\right)^{h\mathbb{F}_p^\times} \simeq (L_{K(n)}\mathbb{S})^{t\Sigma_p},$$

where Σ_p acts trivially [17, Corollary IV.1.4], [13, Lemma 2.1]. On π_0 it is the degree-p power operation followed by the map to Tate cohomology [17, Corollary IV.1.10]. Tate constructions lower chromatic height [12, Section 4]: the target is L_{n-1}-local. The Tate Frobenius therefore factors through a map

$$\psi_n \colon L_{n-1}L_{K(n)}\mathbb{S} \longrightarrow (L_{K(n)}\mathbb{S})^{t\Sigma_p}.$$

For a finite p-local spectrum Y, smashing with Y gives

$$\psi_{n,Y} \colon L_{n-1}L_{K(n)}Y \longrightarrow (L_{K(n)}Y)^{t\Sigma_p}, \tag{1.1}$$

since smashing with a finite spectrum commutes with localizations and with the Tate construction.

Huang proved that $\psi_{n,Y}$ is an equivalence when Y has type $n-1$ [13, Theorem 5.1]. The question is what happens for spectra of smaller type, above all for the sphere. On a type-$(n-1)$ spectrum only the height $n-1$ part of the source survives, so one is led to compare the two sides one chromatic height at a time, and rationally. The following theorem gives a retraction at each positive height and an equivalence after rationalization.

Theorem 1.1. *Let Y be a finite p-local spectrum.*

(i) *The rationalization $L_0\psi_{n,Y}$ is an equivalence.*

(ii) *For every $1 \le t < n$, the map $L_{K(t)}\psi_{n,Y}$ admits a retraction, natural in Y. For $t = n-1$ it is an equivalence.*

(iii) *The map $\psi_{n,Y}$ is an equivalence for $n = 1, 2$. For $n \ge 2$ its cofiber is L_{n-2}-local and rationally zero.*

The theorem is proved by computing the Tate Frobenius on Morava E-theory and descending. On E_n, the Tate Frobenius is the target map of the moduli of degree-p isogenies of the universal deformation ([19, Theorem 1.1], [13, Lemma 3.2]). Inverting the Euler class keeps only the isogenies with étale kernel. Over the height-t locus the étale quotient of the p-divisible group has height $n-t$; its p-torsion is étale locally the constant group $\mathbb{F}_p^{n-t}$. Isogenies with étale kernel ending at a given formal group correspond to hyperplanes in this quotient. There are

$$d_{n,t} = \frac{p^{n-t}-1}{p-1} = 1 + p + \cdots + p^{n-t-1}$$

of them, a number prime to p. The localized Tate Frobenius on each Morava descent term is therefore finite étale of this degree. Its traces descend to a map whose value on the unit is invertible; normalizing by that value gives (ii). For (i) the trace again gives a split injection on rational stabilizer cohomology, and we prove surjectivity by showing that both sides have the same dimension. The isogeny space is an open annulus times a polydisc, and the comparison theorems of [2] together with the two-tower isomorphism compute its rational cohomology as that of an open subgroup of $\mathrm{GL}_n(\mathbb{Z}_p)$. Part (iii) follows formally from (i) and (ii).

The retraction in (ii) need not be an equivalence. Let $\mathbb{S}/p = \mathrm{cofib}(p\colon \mathbb{S} \to \mathbb{S})$.

Theorem 1.2. *At $p = 2$ and $n = 3$,*

$$\pi_0\big(\mathrm{cofib}(L_{K(1)}\psi_3) \wedge \mathbb{S}/2\big) \neq 0.$$

In particular neither ψ_3 nor $\psi_{3,\mathbb{S}/2}$ is an equivalence.

The spectrum $\mathbb{S}/2$ has type one, while Huang's theorem at height three needs type two; so Theorem 1.2 shows that his hypothesis cannot be weakened.

We now explain where the obstruction in Theorem 1.2 comes from. At $n = 3$, $t = 1$ and $p = 2$ the cover of (ii) has degree 3: over the ordinary locus, an isogeny with étale kernel ending at the universal deformation amounts to a line in the rank-two étale Tate module modulo 2. With $a = u_1$ and $b = u_2$, its coordinate ring is the cubic étale algebra

$$k[[U,S]][(US)^{-1}] \quad \text{over } k[[a,b]][a^{-1}], \qquad a = US, \quad b = S^2 + U$$

(Lemma 4.1). The trace splits off the base algebra; write N for its kernel. Its solid coefficient structure is inherited from the completed deformation ring by localization. The modulo-2 coefficient diagram of the cofiber is the continuous bar complex of this solid representation (Proposition 4.4). The problem is to construct a nonzero element in the homotopy of the resulting descent totalization.

The element comes from the next level of structure. A line modulo 2 has two lifts to a line modulo 4, and these form a double cover of the cubic isogeny cover, equivariant for the Morava stabilizer group. Its degree-one stabilizer cohomology class, with coefficients in this cubic algebra,

has nonzero projection to N (Proposition 5.1); this is an explicit Artin–Schreier computation together with the fact that N has no stabilizer invariants. To realize this class in homotopy, we take the stable transfer of the double cover minus two points. Its image in π_0 of the cofiber vanishes on coefficients. Since the degree-one coefficient groups also vanish, this element has a canonical degree-two Postnikov invariant. We compute this invariant by comparison with height-one K-theory. It contains a mixed cup product of the double-cover class and a character of the Adams-operation group (Proposition 7.1). Contracting with the central element $[-1]$ of the stabilizer group kills the square term and sends the mixed term to the nonzero degree-one class above (Section 8).

Conventions

All spectra are p-local, and smash products are derived. For a spectrum M write $M/p = M \wedge \mathbb{S}/p$; for an ordinary ring or module, quotient by (p) has its algebraic meaning. If a ring spectrum A has even, p-torsion-free homotopy, then $\pi_{2j}(A/p) = \pi_{2j}A/(p)$ and $\pi_{2j+1}(A/p) = 0$. Products of these coefficient classes are taken in the algebraic quotient of π_*A. Let Γ_n be the height-n Honda formal group over $k = \mathbb{F}_{p^n}$, and let E_n be the associated Morava E-theory, with

$$\pi_* E_n = W(k)[[u_1, \dots, u_{n-1}]][u^{\pm 1}], \qquad |u| = -2.$$

Write $\mathbb{S}_n = \mathrm{Aut}_k(\Gamma_n)$ for the Morava stabilizer group and $\mathbb{G}_n = \mathbb{S}_n \rtimes \mathrm{Gal}(k/\mathbb{F}_p)$ for the extended one. The completed Amitsur cosimplicial ring spectrum

$$L_{K(n)}(E_n^{\wedge(\bullet+1)})$$

has cofaces induced by inserting the unit and codegeneracies induced by multiplying adjacent factors. Its degree-q term satisfies

$$\pi_* L_{K(n)}(E_n^{\wedge(q+1)}) = \mathrm{Map}_{\mathrm{cts}}(\mathbb{G}_n^q, \pi_* E_n),$$

and its totalization is $L_{K(n)}\mathbb{S}$. By [13, Lemma 3.1], an exact functor Φ on $K(n)$-local spectra gives

$$\Phi(L_{K(n)}\mathbb{S}) \simeq \mathrm{Tot}_q\, \Phi\Big(L_{K(n)}(E_n^{\wedge(q+1)})\Big).$$

We use this for localizations, the Tate construction for the trivial action, and their cofibers. Finite étale algebras over an $\mathbb{E}_\infty$-ring are taken in the sense of [16, Theorem 7.5.0.6]: they correspond to finite étale algebras over π_0, and satisfy $\pi_* B = \pi_* A \otimes_{\pi_0 A} \pi_0 B$ for a finite étale A-algebra B.

We use continuous cohomology in the solid sense of [2, Section 3.3]. For a profinite group G, let Solid^G be the category of solid abelian groups with continuous G-action. A separated, complete linearly topologized module M_0 with continuous G-action defines a solid representation $\underline{M_0}$ [2, Lemmas 3.2.1 and 3.3.1]. Such complete coefficient modules will also be used without an underline when their topology is fixed. Following [2, Notation 3.3.3], for $M \in D(\mathrm{Solid}^G)$ write

$$M^{hG} \in D(\mathrm{Solid}), \qquad H^s(G, M) = H^s(M^{hG}) \in \mathrm{Solid}.$$

Evaluation at the one-point profinite set, denoted by $(*)$, is exact [2, Definition 3.4.4]. Ordinary cohomology classes belong to $H^s(G, M)(*)$. For complete linearly topologized coefficients M_0, the classical continuous cohomology group $H^s_{\mathrm{cts}}(G, M_0)$ identifies with $H^s(G, \underline{M_0})(*)$ [2, Lemma 3.3.4].

Write $\mathbb{Z}[T]^{\blacksquare}$ for the free solid abelian group on a profinite set T. For $M \in \mathrm{Solid}^G$, its scalar continuous bar complex has terms

$$\mathrm{C}^q_{\mathrm{cts}}(G, M) = \mathrm{Hom}_{\mathrm{Solid}}(\mathbb{Z}[G^q]^{\blacksquare}, M),$$

with the inhomogeneous differential; $Z^q_{\mathrm{cts}}(G, M)$ denotes its cocycles. By the solid bar resolution [2, (3.3.5)], this complex computes $(M^{hG})(*)$. For complete linearly topologized coefficients it is the usual continuous cochain complex [2, Lemma 3.3.4].

2 The Tate Frobenius as an isogeny, and retractions

In this section we prove part (ii) of Theorem 1.1. The plan is to describe the Tate Frobenius of E_n as the target map of a moduli space of isogenies, to show that after $K(t)$-localization this map is finite étale of degree $d_{n,t}$, and then to descend its trace to the sphere.

2.1 The ring of subgroups of order p

Let $I_{\mathrm{tr}} \subset E_n^0(B\Sigma_p)$ be the ideal generated by the images of the transfers from $E_n^0(B(\Sigma_i \times \Sigma_{p-i}))$, $1 \le i < p$. Since each proper Young subgroup has order prime to p, the image of its transfer is the image of the transfer from the trivial subgroup. Put

$$\mathcal{O}_{\mathrm{Sub}} = E_n^0(B\Sigma_p)/I_{\mathrm{tr}}.$$

By [19, Theorem 1.1], $\mathcal{O}_{\mathrm{Sub}}$ is the coordinate ring of the scheme of subgroups of order p of the universal deformation $\mathcal{G}$ over $\pi_0 E_n$. Thus it carries two maps from $\pi_0 E_n$: the *source* map, the $\pi_0 E_n$-algebra structure, which remembers $\mathcal{G}$, and the *target* map, which remembers the quotient $\mathcal{G}/K$ by the universal subgroup K together with its induced deformation structure. Huang shows that the Tate Frobenius $\pi_0 E_n \to \pi_0(E_n^{t\Sigma_p})$ is the target map followed by inverting the Euler class of K [13, Lemma 3.2]. Both maps are finite flat of degree $(p^n - 1)/(p - 1)$ [18, Proposition 13.1].

To study the target map we need a description of $\mathcal{O}_{\mathrm{Sub}}$ as a moduli space over the *target*. Let σ be the Frobenius of k; the closed fiber of the universal isogeny is the relative Frobenius $F\colon \Gamma_n \to \Gamma_n^\sigma$, so the target is a deformation of Γ_n^σ. Write $\mathcal{G}'$ for the universal deformation of Γ_n^σ.

Lemma 2.1. *Over the deformation ring of Γ_n^σ, the target map identifies* $\operatorname{Spec}\mathcal{O}_{\mathrm{Sub}}$ *with the scheme of finite locally free subgroups $H \subset \mathcal{G}'[p]$ of rank p^{n-1}. A subgroup H corresponds to the degree-p isogeny $\mathcal{G}'/H \to \mathcal{G}'$ induced by $[p]$. This identification commutes with base change and with isomorphisms of the target.*

Proof. The idea is that a degree-p isogeny q ending at $\mathcal{G}'$ and the subgroup H determine each other through the factorization

$$[p]\colon \mathcal{G}' \longrightarrow \mathcal{G}'/H \xrightarrow{q} \mathcal{G}'.$$

Given H, the map $[p]$ kills H and so factors as displayed, and q has degree $p^n/p^{n-1} = p$. Conversely, given q of degree p, its kernel is a finite locally free group of order p, hence killed by p by Deligne's theorem [21, §1], so $[p]$ factors through q and the kernel of the complementary isogeny is a subgroup H of rank p^{n-1}. The two constructions are inverse to each other and commute with base change.

It remains to see that the subgroup scheme is finite over the target deformation ring, so that it can be compared with $\mathcal{O}_{\mathrm{Sub}}$. It is a closed subscheme of a relative Grassmannian of $\mathcal{G}'[p]$,

hence proper. Its closed fiber is one point, since the only ideal of colength p^{n-1} in $k[x]/x^{p^n}$ is $(x^{p^{n-1}})$. The locus of positive-dimensional fibers is closed, and would contain the closed point if it were nonempty; hence the scheme is quasi-finite and proper, so finite. On the closed fiber the subgroup is $\ker F^{n-1}$, with quotient $\Gamma_n^\sigma/\ker F^{n-1} = \Gamma_n^{\sigma^n} = \Gamma_n$, and the complementary isogeny is $F\colon \Gamma_n \to \Gamma_n^\sigma$. So the two constructions identify the deformation functors with these framings on Artin local rings. Both coordinate rings are finite and complete over the target deformation ring, so they are isomorphic. □

2.2 The localized Tate Frobenius is finite étale

Proposition 2.2. *Let $n \geq 2$ and $1 \leq t < n$. The Tate Frobenius*

$$\psi\colon L_{K(t)}E_n \longrightarrow L_{K(t)}(E_n^{t\Sigma_p})$$

is a finite étale map of even-periodic $\mathbb{E}_\infty$-rings, of degree $d_{n,t} = (p^{n-t}-1)/(p-1)$. For every $q \geq 0$, the natural map

$$\begin{aligned} L_{K(t)}L_{K(n)}(E_n^{\wedge(q+1)}) \otimes_{L_{K(t)}E_n,\psi} L_{K(t)}(E_n^{t\Sigma_p}) \\ \longrightarrow L_{K(t)}\Big((L_{K(n)}(E_n^{\wedge(q+1)}))^{t\Sigma_p}\Big) \end{aligned} \tag{2.1}$$

is an equivalence, and these equivalences are compatible with all cosimplicial structure maps.

Proof. The point is that inverting the Euler class keeps exactly the isogenies with étale kernel, and over the height-t locus these are counted by hyperplanes in an $\mathbb{F}_p$-vector space of dimension $n-t$.

Euler class and étale kernels. Let q be the universal isogeny of degree p ending at $\mathcal{G}'$, with kernel K. After passing to the finite flat cover on which K acquires a generator s, the Euler class of K is $\prod_{j=1}^{p-1} x([j]s)$, where x is a coordinate. It is a unit exactly when the sections $[j]s$, $j \neq 0$, are disjoint from the zero section, that is, when K is étale. The condition is invariant under the action of $\mathbb{F}_p^\times$ on generators, so it descends. Hence the localized Tate Frobenius is the target map restricted to the open locus where K is étale.

The height-t locus. Put $I_t = (p, u_1, \dots, u_{t-1})$. Modulo I_t and with u_t inverted, the p-divisible group of $\mathcal{G}'$ is an extension of an étale group ${\mathcal{G}'}^{\mathrm{et}}$ of height $n-t$ by a connected group of height t [20, §2]. In the description of Lemma 2.1, $K = \mathcal{G}'[p]/H$, so K is étale exactly when H contains the connected part ${\mathcal{G}'}^0[p]$. Such H are the inverse images of hyperplanes in ${\mathcal{G}'}^{\mathrm{et}}[p]$, which is étale locally $\mathbb{F}_p^{n-t}$. They form a finite étale scheme whose geometric fibers are $\mathbb{P}^{n-t-1}(\mathbb{F}_p)$, with $d_{n,t}$ points. Since we take inverse images in $\mathcal{G}'[p]$, this does not require the connected–étale sequence to split.

The completed étale branch. The localized coefficient ring is

$$\pi_0 L_{K(t)}E_n = \big((\pi_0 E_n)[u_t^{-1}]\big)^\wedge_{I_t}.$$

Form the finite flat algebra

$$\mathcal{O}^{(t)}_{\mathrm{Sub}} = \mathcal{O}_{\mathrm{Sub}} \otimes_{\pi_0 E_n,\mathrm{tgt}} \pi_0 L_{K(t)}E_n.$$

It is I_t-adically complete. Let e_K denote the Euler class used above. In the special fiber $\mathcal{O}^{(t)}_{\mathrm{Sub}}/I_t$, the open locus where e_K is invertible is the finite étale scheme of hyperplanes just described. Since it is finite over the base, this open locus is also closed. Its idempotent lifts uniquely to

$\mathcal{O}^{(t)}_{\mathrm{Sub}}$, giving a product decomposition. On the selected factor e_K is a unit. On the other factor it is nilpotent modulo I_t, so inverting it makes the I_t-adic completion zero. Consequently

$$(\mathcal{O}^{(t)}_{\mathrm{Sub}}[e_K^{-1}])^{\wedge}_{I_t}$$

is exactly the selected finite flat factor. Its special fiber is étale, so the factor itself is finite étale over $\pi_0 L_{K(t)} E_n$, of rank $d_{n,t}$.

It remains to identify this algebra with the actual localized Tate coefficients. On the Euler-invertible locus the isogeny q has invertible derivative, hence induces an isomorphism of the formal completions at the identity of its source and target. The ideals $(p, u_1, \dots, u_{i-1})$ are the invariant height ideals of these formal groups: a change of coordinate identifies the two ideals, and identifies the first surviving coefficient u_i up to a unit modulo that ideal. Thus, after inverting e_K, localizing at u_t and completing at I_t give the same algebra whether performed through the source or the target map. The coefficient formula for localization of a complex-oriented module [4, Proposition 3.7], applied to $\pi_*(E_n^{t\Sigma_p}) = \mathcal{O}_{\mathrm{Sub}}[e_K^{-1}][u^{\pm 1}]$, now identifies $\pi_0 L_{K(t)}(E_n^{t\Sigma_p})$ with the displayed completed factor. The sequence defining I_t is regular on the finite flat target algebra; its quotients have surjective completion transition maps. Hence the derived completion agrees with the ordinary completion and introduces no odd homotopy. Periodicity is preserved as well. The map on all homotopy groups is therefore the even-periodic base change of this finite étale map on π_0. By [16, Theorem 7.5.0.6], the given map of $\mathbb{E}_\infty$-rings is finite étale.

The Amitsur complex. Huang's formula [13, Lemma 4.1, (3)] gives

$$\big(L_{K(n)}(E_n^{\wedge(q+1)})\big)^{t\Sigma_p} \simeq L_{K(n)}(E_n^{\wedge(q+1)}) \otimes_{E_n,\mathrm{src}} E_n^{t\Sigma_p}$$

for the source structure: its coefficients are continuous functions on $\mathbb{G}_n^q$ with values in the isogeny ring, with the Euler class inverted. The source and target maps of the isogeny ring have the same closed point, so the two adic topologies agree. The evaluation maps $L_{K(n)}(E_n^{\wedge(q+1)}) \to E_n$ are maps of $\mathbb{E}_\infty$-rings, so by naturality the Tate Frobenius sends a function f to $g \mapsto \psi(f(g))$. On the étale branch the isogeny algebra is finite projective over the target. Choose an idempotent presentation as a summand of a finite free module. The idempotent and its two comparison maps involve finitely many coefficients. Modulo each I_t^j these coefficients have a common power of u_t as denominator, with a bound that may depend on j. The idempotent presentation therefore identifies the localized continuous function modules with functions in the localized finite projective module having a common denominator at each fixed j. Taking the inverse limit in j gives (2.1) on homotopy groups, hence on spectra.

The action coface on the target coefficient algebra transports a subgroup along an isomorphism of its target formal group. The complementary isogeny construction in Lemma 2.1 commutes with that transport. On the closed fiber this uses $F \circ g = g^\sigma \circ F$, so the induced target framing is the Frobenius-twisted framing fixed in that lemma. The remaining cofaces compose these isomorphisms, and the codegeneracies insert identities. Thus the pointwise Tate Frobenius commutes with every coface and codegeneracy. The idempotent selection and completion above are functorial for these maps, proving compatibility of the whole cosimplicial diagrams. □

Proposition 2.3. *Let $n \geq 2$, $1 \leq t < n$, and Y a finite p-local spectrum. The map*

$$L_{K(t)}\psi_{n,Y} \colon L_{K(t)}L_{K(n)}Y \longrightarrow L_{K(t)}\big((L_{K(n)}Y)^{t\Sigma_p}\big)$$

admits a retraction, natural in Y. For $t = n-1$ it is an equivalence.

Proof. It suffices to treat $Y = \mathbb{S}$ and smash with Y, since Y is finite. A finite étale map of $\mathbb{E}_\infty$-rings has a trace, a module map back to the base whose composite with the unit is multiplication by the trace of 1. By Proposition 2.2, the localized Tate Frobenius maps on the completed Amitsur diagram are finite étale. The base-change equivalences (2.1) make their traces compatible with the cosimplicial structure maps. These traces totalize, by [13, Lemma 3.1], to a map of $L_{K(t)}L_{K(n)}\mathbb{S}$-modules

$$\mathrm{Tr}\colon L_{K(t)}\big((L_{K(n)}\mathbb{S})^{t\Sigma_p}\big) \longrightarrow L_{K(t)}L_{K(n)}\mathbb{S}.$$

The composite $\mathrm{Tr}\circ L_{K(t)}\psi_n$ is multiplication by $e = \mathrm{Tr}(1)$. In each term of the Amitsur complex, e maps to $d_{n,t}$, which is a unit in $\mathbb{Z}_p$. Multiplication by e is therefore an equivalence termwise, hence after totalization, and $e^{-1}\,\mathrm{Tr}$ is a retraction. If $t = n-1$ then $d_{n,t} = 1$, the maps of the Amitsur complexes are equivalences, and so is their totalization. □

3 The rational comparison

We prove part (i) of Theorem 1.1, and deduce part (iii).

Proposition 3.1. *For every prime p and every $n \geq 1$, the map*

$$L_0\psi_n\colon L_0L_{K(n)}\mathbb{S} \longrightarrow L_0\big((L_{K(n)}\mathbb{S})^{t\Sigma_p}\big)$$

is an equivalence, and the same holds after smashing with any finite p-local spectrum. In particular $\pi_*L_0((L_{K(n)}\mathbb{S})^{t\Sigma_p}) \cong \Lambda_{\mathbb{Q}_p}(\zeta_1,\dots,\zeta_n)$ with $|\zeta_i| = 1-2i$.

Throughout this section, rational continuous cohomology means the cohomology of the integral continuous cochain complex followed by inversion of p:

$$H^s_{\mathrm{cts}}(G,M)[1/p] = H^s\big((M^{hG}[1/p])(*)\big).$$

In particular, this notation does not introduce a separate topology on $M[1/p]$. Homotopy fixed point comparisons take place in $D(\mathrm{Solid})$; dimensions refer to their scalar cohomology. The two sides of the proposition are computed by termwise rationalization of their integral Amitsur descent diagrams. The proof compares the resulting cochain complexes. A central element of $\mathbb{G}_n$ kills all periodic degrees except degree zero, so only π_0 matters. In degree zero, the trace of the finite flat target map $\pi_0E_n \to \mathcal{O}_{\mathrm{Sub}}$ shows that

$$H^*_{\mathrm{cts}}(\mathbb{G}_n,\pi_0E_n)[1/p] \longrightarrow H^*_{\mathrm{cts}}(\mathbb{G}_n,\mathcal{O}_{\mathrm{Sub}})[1/p]$$

is split injective, and it remains to show that the two sides have the same finite dimension in each degree. The left side is known [2, Theorem B]. For the right side we use that $\mathcal{O}_{\mathrm{Sub}}$ is the coordinate ring of an annulus times a polydisc (Lemma 3.2) and that, on such a space, the integral comparison theorem of [2] holds with a uniform error (Lemma 3.3). The two-tower isomorphism then computes the right side as the cohomology of an open subgroup of $\mathrm{GL}_n(\mathbb{Z}_p)$.

3.1 The isogeny space is an annulus times a polydisc

The structure sheaves in the following comparisons carry their condensed enhancements, as in [2, Section 3.5]; derived global sections and continuous derived invariants are taken with this structure.

Lemma 3.2. *Let $n \geq 2$. There are elements $\xi, \eta \in \mathcal{O}_{\mathrm{Sub}}$ such that*

$$\mathcal{O}_{\mathrm{Sub}} \cong W(k)[[\xi, \eta, u_2, \ldots, u_{n-1}]]/(\xi\eta + p), \tag{3.1}$$

and the Euler class of the universal subgroup is a unit multiple of ξ.

Proof. Choose a p-typical coordinate z on $\mathcal{G}$ on which the Teichmüller roots of unity $\mu_{p-1} \subset \mathbb{Z}_p^\times$ act linearly, $[\zeta](z) = \zeta z$. Then $[p](z)/z = Q(z^{p-1})$ for a power series Q with $Q(0) = p$, and the description of $\mathcal{O}_{\mathrm{Sub}}$ as the ring of subgroups gives $\mathcal{O}_{\mathrm{Sub}} = (\pi_0 E_n)[[\xi]]/(Q(\xi))$ with $\xi = z^{p-1}$. Put $\eta = (Q(\xi) - p)/\xi$, so that $\xi\eta + p = Q(\xi) = 0$ in $\mathcal{O}_{\mathrm{Sub}}$.

We show that η can replace u_1 as a parameter. Write the logarithm as $\log(z) = z + (u_1/p)z^p + \cdots$. Comparing coefficients of z^p in $\log([p](z)) = p\log(z)$ shows that $[p](z) = pz + (1 - p^{p-1})u_1 z^p + \cdots$. Hence the linear coefficient of Q is $(1 - p^{p-1})u_1$, and

$$\eta \equiv (1 - p^{p-1})u_1 \pmod{\xi}.$$

So $\partial\eta/\partial u_1$ is the unit $1 - p^{p-1}$ at the closed point. In $(\pi_0 E_n)[[\xi]] = W(k)[[\xi, u_1, \ldots, u_{n-1}]]$ the formal inverse function theorem lets us replace u_1 by η, and the relation $Q(\xi) = 0$ becomes $\xi\eta + p = 0$. This is (3.1). Finally the Euler class is the product of the coordinates of the nonzero multiples $[\zeta]z = \zeta z$ of a generator, $\prod_{\zeta \in \mu_{p-1}} \zeta z$, a unit multiple of $z^{p-1} = \xi$. □

The generic fiber of $\operatorname{Spf} \mathcal{O}_{\mathrm{Sub}}$ is thus the product of the open annulus $|p| < |\xi| < 1$ (on which $\eta = -p/\xi$) and an open ball of dimension $n - 2$. For a complex M of $\mathbb{Z}_p$-modules write $M[\epsilon] = M \otimes \Lambda_{\mathbb{Z}_p}(\epsilon)$ with ϵ in cohomological degree one. As in [2, Section 5.3], ϵ is the arithmetic class in the integral comparison.

Lemma 3.3. *Let $d \geq 0$ be an integer and let $\mathcal{O}$ be a complete discrete valuation ring of mixed characteristic $(0, p)$ with perfect residue field. Put $B = \mathcal{O}[[\xi, \eta, T_1, \ldots, T_d]]/(\xi\eta + p)$, and let X be the generic fiber of* $\operatorname{Spf} B$ *for the maximal-ideal-adic topology. Regard B with this topology as a condensed abelian group. There is a natural morphism in the derived category of condensed abelian groups*

$$B[\epsilon] \longrightarrow R\Gamma(X_{\mathrm{pro\acute{e}t}}, \widehat{\mathcal{O}}^+)$$

whose cofiber has cohomology killed by p^c for a constant c independent of the degree. The morphism is equivariant for all automorphisms of $\operatorname{Spf} B$*, including semilinear ones, acting on $B[\epsilon]$ through B and fixing the arithmetic class ϵ.*

Proof. The idea is to exhaust X by affinoids that have semistable models, apply the comparison of [2] on each with an error bound that does not depend on the affinoid, and pass to the limit. The only new point is that the bounded functions on the exhaustion recover B without higher derived limits.

The exhaustion. For primes $\ell > 2$ different from p, let $X_\ell \subset X$ be the affinoid $|\xi|, |\eta|, |T_i| \leq |p|^{1/\ell}$; these exhaust X as $\ell \to \infty$. After a finite tamely ramified Galois extension L of the fraction field of $\mathcal{O}$ containing $p^{1/\ell}$, put $\xi = p^{1/\ell}U'$, $\eta = p^{1/\ell}V'$ and $T_i = p^{1/\ell}W_i$; then X_ℓ has the model

$$\mathcal{O}_L\langle U', V', W_1, \ldots, W_d\rangle/(U'V' + p^{1-2/\ell}).$$

This is a semistable chart in the sense of [2, Definition 5.2.1], where the parameter is allowed any positive rational valuation. By [2, Lemmas 5.6.1–5.6.2], the comparison map $\mathcal{O}^+(X_\ell)[\epsilon] \to R\Gamma(X_{\ell,\mathrm{pro\acute{e}t}}, \widehat{\mathcal{O}}^+)$ has cofiber cohomology killed by p^N in every degree, and by [2, Theorem 5.3.4] the constant N depends only on the ground field and on the dimension, not on ℓ.

The limit of bounded functions. We claim that $R\varprojlim_\ell \mathcal{O}^+(X_\ell) \simeq B$. Every element of $\mathcal{O}^+(X_\ell)$ is a convergent sum of the monomials $\xi^a T^I$ $(a \geq 0)$ and $\eta^b T^I$ $(b \geq 1)$, and a monomial of total degree D has sup norm $|p|^{D/\ell}$ on X_ℓ. Fix ℓ_0 and $j \geq 1$. For $\ell \geq 2\ell_0$, a term of degree D in an element of $\mathcal{O}^+(X_\ell)$ has coefficient of absolute value at most $|p|^{-D/\ell}$, so on X_{ℓ_0} it has norm at most $|p|^{D(1/\ell_0 - 1/\ell)} \leq |p|^{D/(2\ell_0)}$. Hence modulo p^j only the terms of degree $D < 2j\ell_0$ survive on X_{ℓ_0}, and their coefficients range over a set that stabilizes as ℓ grows, because the valuation is discrete. So the images of $\mathcal{O}^+(X_\ell)/p^j \to \mathcal{O}^+(X_{\ell_0})/p^j$ stabilize, the system is Mittag-Leffler, and the same holds after evaluation on a profinite set [2, Lemma 6.1.2]. Each $\mathcal{O}^+(X_\ell)$ is derived p-complete, so

$$R\varprojlim_\ell \mathcal{O}^+(X_\ell) \simeq R\varprojlim_{\ell,j} \mathcal{O}^+(X_\ell)/p^j \simeq \varprojlim_{\ell,j} \mathcal{O}^+(X_\ell)/p^j.$$

The last limit consists of the series whose coefficients are bounded on every X_ℓ, which by the norm computation is B; its limit topology is the $(p, \xi, \eta, T_1, \ldots, T_d)$-adic one.

Conclusion. Taking $R\varprojlim_\ell$ of the comparisons on X_ℓ gives the map. Since a countable inverse limit has cohomological dimension one, the cofiber cohomology is killed by p^{2N}. The comparisons on X_ℓ are natural, so the limit map depends only on X and is equivariant for its automorphisms. □

3.2 The finite-level two-tower comparison

Let X_n be the generic fiber of $\operatorname{Spf} \mathcal{O}_{\mathrm{Sub}}$, viewed over the deformation space of the target by Lemma 2.1. Let H be Drinfeld's symmetric space over $\mathbb{Q}_p$, the complement in $\mathbb{P}^{n-1}_{\mathbb{Q}_p}$ of all $\mathbb{Q}_p$-rational hyperplanes. Put

$$P_n = \{g \in \mathrm{GL}_n(\mathbb{Z}_p) : \overline{g}(\mathbb{F}_p^{n-1} \times 0) = \mathbb{F}_p^{n-1} \times 0\}.$$

Thus P_n is the inverse image of a hyperplane stabilizer under reduction modulo p.

Lemma 3.4. *For $n \geq 2$ there is an equivalence of v-stacks over $\mathbb{Q}_p$*

$$[X_n^\diamond / \mathbb{G}_n] \simeq [H^\diamond / P_n]. \tag{3.2}$$

Here $\mathbb{G}_n$ acts through transport of the target formal group, and P_n acts through the usual $\mathrm{GL}_n(\mathbb{Q}_p)$-action on H. It induces an equivalence in $D(\mathrm{Solid})$

$$R\Gamma(X_{n,\mathrm{pro\acute{e}t}}, \widehat{\mathcal{O}}^+)^{h\mathbb{G}_n} \simeq R\Gamma(H_{\mathrm{pro\acute{e}t}}, \widehat{\mathcal{O}}^+)^{hP_n}. \tag{3.3}$$

Proof. We first make the unramified descent explicit. Let $\overline{\mathrm{LT}}_n$ be the target Lubin–Tate deformation space over $W(\overline{\mathbb{F}}_p)[1/p]$, and let $\overline{X}_n$ be the generic fiber of

$$\overline{\mathcal{O}_{\mathrm{Sub}}} = \mathcal{O}_{\mathrm{Sub}} \widehat{\otimes}_{W(k),\,\mathrm{tgt}} W(\overline{\mathbb{F}}_p).$$

The completion is for the deformation ideal; in the coordinates of Lemma 3.2 its ring is $W(\overline{\mathbb{F}}_p)[[\xi, \eta, u_2, \ldots, u_{n-1}]]/(\xi\eta + p)$. Put $\Delta_n = \mathrm{Gal}(\overline{\mathbb{F}}_p/k)$. The standard Honda model over $\mathbb{F}_p$ gives an action-compatible exact sequence

$$1 \longrightarrow \Delta_n \longrightarrow \widetilde{\mathbb{G}}_n \longrightarrow \mathbb{G}_n \longrightarrow 1, \qquad \widetilde{\mathbb{G}}_n = \mathbb{S}_n \rtimes \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p).$$

All endomorphisms of the Honda group are defined over k, so the action on $\mathbb{S}_n$ factors through $\mathrm{Gal}(k/\mathbb{F}_p)$, and Δ_n acts on $\overline{\mathcal{O}_{\mathrm{Sub}}}$ only through its Witt coefficients. This $\widetilde{\mathbb{G}}_n$ is the group denoted by G_n in [2, Section 3.9].

The map $\overline{X}_n^\diamond \to X_n^\diamond$ is a pro-étale Δ_n-torsor. Indeed, it is the completed inverse limit of the finite unramified coefficient extensions. The finite Galois torsor identities, first modulo each power of the deformation ideal and then after completion, give its Čech relation

$$\overline{X}_n^\diamond \times_{X_n^\diamond} \overline{X}_n^\diamond \cong \overline{X}_n^\diamond \times \underline{\Delta_n}.$$

It is surjective: an algebraically closed complete geometric point admits a compatible embedding of the completed maximal unramified coefficient field. Consequently quotienting in stages gives

$$[X_n^\diamond/\mathbb{G}_n] \simeq [\overline{X}_n^\diamond/\widetilde{\mathbb{G}}_n]. \tag{3.4}$$

We next identify the finite level in the two-tower theorem. Let M_n be the v-stack of one-dimensional formal groups of height n. The construction of [3, Theorem 3.6.1] identifies $H^\diamond$ with the torsor over M_n of frames of the universal Tate local system. Indeed, the local system in its shtuka description is the actual Tate module, by [3, Theorem 3.5.3 and the following paragraph], and the proof of Theorem 3.6.1 obtains $H^\diamond$ by trivializing it. Define

$$\mathrm{LT}_{n,\infty} = \overline{\mathrm{LT}}_n^\diamond \times_{M_n} H^\diamond.$$

This is the common perfectoid space of [2, Theorem 3.9.1 and the following torsor diagram]:

$$\begin{array}{ccc} & \mathrm{LT}_{n,\infty} & \\ {\scriptstyle \mathrm{GL}_n(\mathbb{Z}_p)}\swarrow & & \searrow{\scriptstyle \widetilde{\mathbb{G}}_n} \\ \overline{\mathrm{LT}}_n^\diamond & & H^\diamond. \end{array}$$

Each arrow is the base change of the other atlas over M_n, hence is a pro-étale torsor for the indicated group; the two actions commute. The left arrow is the torsor of frames $\beta\colon \underline{\mathbb{Z}_p^n} \xrightarrow{\sim} T_p(\mathcal{G}')$ of the target Tate local system. These are identifications of sheaves on perfectoid test spaces. The Honda model over $\mathbb{F}_p$ allows the same construction for the target twist Γ_n^σ of Lemma 2.1.

A frame modulo P_n remembers exactly the hyperplane $\overline{\beta}(\mathbb{F}_p^{n-1} \times 0)$ in $T_p(\mathcal{G}')/p$, where $\overline{\beta}$ is its reduction modulo p. By Lemma 2.1, the finite cover over the target deformation ring represents subgroups of rank p^{n-1} in $\mathcal{G}'[p]$. On the characteristic-zero generic fiber, $\mathcal{G}'[p]$ is finite étale and is naturally $T_p(\mathcal{G}')/p$, so this is precisely its hyperplane local system. Since finite morphisms commute with passage to the adic generic fiber, the moduli identification gives a $\widetilde{\mathbb{G}}_n$-equivariant isomorphism

$$\mathrm{LT}_{n,\infty}/P_n \cong \overline{X}_n^\diamond.$$

The right action on frames is $\beta \cdot g = \beta \circ g$; on H it is $h \cdot g = g^{-1}h$. Its right quotient is therefore the usual left quotient $[H^\diamond/P_n]$. Taking the two commuting quotients now gives

$$[\overline{X}_n^\diamond/\widetilde{\mathbb{G}}_n] \simeq [\mathrm{LT}_{n,\infty}/(\widetilde{\mathbb{G}}_n \times P_n)] \simeq [H^\diamond/P_n].$$

Together with (3.4) this proves (3.2).

For the cohomological assertion, use the pro-étale torsor $\mathrm{LT}_{n,\infty} \to \overline{X}_n$ with group P_n, the torsor $\mathrm{LT}_{n,\infty} \to H$ with group $\widetilde{\mathbb{G}}_n$, and unramified descent to X_n. For a torsor $Y \to Z$ with profinite group K, its Čech nerve has terms $Y \times \underline{K}^q$; the associated complex of derived sections of $\widehat{\mathcal{O}}^+$ is the continuous bar complex. This is the descent argument of [2, Proposition 3.6.3]. It identifies both sides of (3.3) with

$$R\Gamma(\mathrm{LT}_{n,\infty,\mathrm{pro\acute{e}t}}, \widehat{\mathcal{O}}^+)^{h(\widetilde{\mathbb{G}}_n \times P_n)}.$$

The identifications are induced by pullback of the same structure sheaf and commute with pullback from the arithmetic base $\mathbb{Q}_p$. □

3.3 Proof of the rational comparison

Proof of Proposition 3.1. For $n = 1$, apply [13, Theorem 5.1] to the type-zero spectrum $\mathbb{S}$; its comparison is already an equivalence. Let $n \geq 2$.

Reduction to degree zero. Let $\lambda \in 1 + p^2\mathbb{Z}_p$ be non-torsion, viewed as a central element of $\mathbb{G}_n$. It acts trivially on $\pi_0 E_n$ and on the moduli of subgroups, hence on $\mathcal{O}_{\mathrm{Sub}}$, and it acts on $\pi_{2j}E_n$ and $\pi_{2j}E_n^{t\Sigma_p}$ by λ^j. For $j \neq 0$, $\lambda^j - 1$ is invertible in $\mathbb{Q}_p$, and since a central element acts on continuous cohomology as the identity [2, Lemma 2.6.1], the rational continuous cohomology of these twisted modules vanishes. So both descent spectral sequences are concentrated in the row of π_0.

The trace. The target map $\pi_0 E_n \to \mathcal{O}_{\mathrm{Sub}}$ is finite flat of degree $(p^n - 1)/(p - 1)$ and $\mathbb{G}_n$-equivariant; its trace is continuous and equivariant, and its composite with the target map is multiplication by that degree. So the induced map on rational continuous cohomology is split injective.

Coefficient-field descent. Use $\overline{\mathcal{O}_{\mathrm{Sub}}}$, Δ_n and $\widetilde{\mathbb{G}}_n$ from the proof of Lemma 3.4. Let φ be the Witt Frobenius on $W(\overline{\mathbb{F}}_p)$. On the base-changed nodal presentation of Lemma 3.2, a topological generator of Δ_n acts as φ^n on Witt coefficients and fixes the formal variables. The sequence

$$0 \longrightarrow \mathcal{O}_{\mathrm{Sub}} \longrightarrow \overline{\mathcal{O}_{\mathrm{Sub}}} \xrightarrow{\varphi^n - 1} \overline{\mathcal{O}_{\mathrm{Sub}}} \longrightarrow 0$$

is exact: modulo p this is coefficientwise surjectivity of $x \mapsto x^{p^n} - x$ on $\overline{\mathbb{F}}_p$; successive Witt coordinates give surjectivity modulo every p^j, and completion gives the stated sequence. The same argument modulo powers of the maximal ideal, followed by their surjective inverse limit, computes continuous Δ_n-cohomology: its invariants are $\mathcal{O}_{\mathrm{Sub}}$ and its higher cohomology vanishes, as in [2, Lemma 3.8.5(1)]. Hence

$$\overline{\mathcal{O}_{\mathrm{Sub}}}^{h\widetilde{\mathbb{G}}_n} \simeq \mathcal{O}_{\mathrm{Sub}}^{h\mathbb{G}_n}.$$

With $\pi_0 E_n$ in place of $\mathcal{O}_{\mathrm{Sub}}$, the same calculation transports [2, Theorem B] to our finite-residue-field convention.

Equivariant integral comparisons. Lemma 3.3 and [2, Theorem 6.2.6] give respectively the equivariant morphisms

$$\mathcal{O}_{\mathrm{Sub}}[\epsilon] \longrightarrow R\Gamma(X_{n,\mathrm{pro\acute{e}t}}, \widehat{\mathcal{O}}^+),$$
$$\mathbb{Z}_p[\epsilon] \longrightarrow R\Gamma(H_{\mathrm{pro\acute{e}t}}, \widehat{\mathcal{O}}^+).$$

Their cofibers are bounded below, with cohomology killed by a fixed power of p. This suffices to take homotopy fixed points before inverting p: in the hypercohomology spectral sequence of either cofiber, each fixed total degree involves finitely many rows, each killed by that power of p. Its resulting cohomology in that degree is therefore killed by a power of p as well. No uniform bound on the degrees of the resulting group cohomology is needed here. Taking $\mathbb{G}_n$- and P_n-homotopy fixed points, respectively, and using Lemma 3.4 therefore gives

$$\begin{aligned}(\mathcal{O}_{\mathrm{Sub}}^{h\mathbb{G}_n}[\epsilon])[1/p] &\simeq R\Gamma([X_n^\diamond/\mathbb{G}_n]_{\mathrm{pro\acute{e}t}}, \widehat{\mathcal{O}}^+)[1/p] \\ &\simeq (\mathbb{Z}_p^{hP_n}[\epsilon])[1/p].\end{aligned}$$

The arithmetic generator ϵ is the same in both comparisons. Passing to cohomology yields

$$H^*_{\mathrm{cts}}(\mathbb{G}_n, \mathcal{O}_{\mathrm{Sub}})[1/p] \otimes \Lambda_{\mathbb{Q}_p}(\epsilon) \cong H^*_{\mathrm{cts}}(P_n, \mathbb{Q}_p) \otimes \Lambda_{\mathbb{Q}_p}(\epsilon). \tag{3.5}$$

Since P_n is open in $\mathrm{GL}_n(\mathbb{Z}_p)$, Lazard's theorem identifies $H^*_{\mathrm{cts}}(P_n, \mathbb{Q}_p)$ with the invariant Lie algebra cohomology of $\mathfrak{gl}_n$, an exterior algebra on generators of degrees $1, 3, \dots, 2n-1$; the finite group $P_n/(P_n \cap (1+p^2 M_n(\mathbb{Z}_p)))$ acts trivially on it because it acts through inner automorphisms. The same exterior algebra is $H^*_{\mathrm{cts}}(\mathbb{G}_n, \pi_0 E_n)[1/p]$ by [2, Theorem B]. In degree s, the isomorphism (3.5) relates the sum of the dimensions in degrees s and $s-1$ on each side, so by induction on s the groups $H^s_{\mathrm{cts}}(\mathbb{G}_n, \mathcal{O}_{\mathrm{Sub}})[1/p]$ and $H^s_{\mathrm{cts}}(\mathbb{G}_n, \pi_0 E_n)[1/p]$ have the same finite dimension. The split injection is therefore an isomorphism.

Descent. In $\mathcal{O}_{\mathrm{Sub}}$ we have $p = -\xi\eta$, so inverting p inverts the Euler class ξ (up to a unit). More precisely, in every Amitsur degree the coefficient ring before rationalization is the continuous-function module with values in $\mathcal{O}_{\mathrm{Sub}}$, localized at that Euler class. Rationalization therefore identifies it with the same integral continuous-function module followed by inversion of p. Thus the integral-first cohomology computed above is the actual E_2-term of the rationalized Tate descent diagram. Applying [13, Lemma 3.1] to rationalization and to rationalized Tate, the two descent spectral sequences are compared by ψ; we have shown that the map of E_2-terms is an isomorphism, and both are concentrated in a finite range of cohomological degrees. Descendability makes the totalization towers pro-constant, so the spectral sequences converge and $L_0\psi_n$ is an equivalence. The description of the homotopy is [2, Theorem A]. For a finite spectrum Y, smash with Y. □

Corollary 3.5. *For every finite p-local Y and $n \geq 2$, the cofiber of $\psi_{n,Y}$ is L_{n-2}-local and rationally zero. In particular $\psi_{n,Y}$ is an equivalence for $n = 1, 2$.*

Proof. Both sides of $\psi_{n,Y}$ are L_{n-1}-local, the target by [12, §4]. By Proposition 2.3 with $t = n-1$ the cofiber is $K(n-1)$-acyclic, hence L_{n-2}-local by the chromatic fracture square, and by Proposition 3.1 it is rationally zero. For $n = 2$ an L_0-local spectrum with zero rationalization is zero. The case $n = 1$ is [13, Theorem 5.1]. □

Theorem 1.1 is now proved: (i) is Proposition 3.1, (ii) is Proposition 2.3, and (iii) is Corollary 3.5.

4 Height three: reduction to a descent problem

From now on $p = 2$ and $n = 3$, so $k = \mathbb{F}_8$. Choose the deformation parameters of [11, (3.3.13) and Example 3.3.18], and write $a = u_1$ and $b = u_2$, so that $\pi_* E_3 = W(k)[[a, b]][u^{\pm 1}]$, and $v_1 = au^{-1} \in \pi_2 E_3$; modulo 2 the class v_1 is invariant under $\mathbb{G}_3$. Let $C^\bullet$ be the cofiber of the localized Tate Frobenius on the completed Morava resolution; thus

$$C^q = \mathrm{cofib}\Big(L_{K(1)} L_{K(3)} (E_3^{\wedge(q+1)}) \xrightarrow{\psi} L_{K(1)} \big((L_{K(3)} (E_3^{\wedge(q+1)}))^{tC_2} \big) \Big). \tag{4.1}$$

Since $K(1)$-localization, the Tate construction for the trivial action and cofibers are exact, and $\mathbb{S}/2$ is finite, [13, Lemma 3.1] gives

$$\mathrm{Tot}(C^\bullet/2) \simeq \mathrm{cofib}(L_{K(1)}\psi_3) \wedge \mathbb{S}/2. \tag{4.2}$$

Theorem 1.2 therefore asks for a nonzero element of $\pi_0 \mathrm{Tot}(C^\bullet/2)$. In this section we compute the coefficients of the two terms in (4.1) and of $C^q/2$; in Section 5 we find a nonzero class in the cohomology of these coefficients, and in Sections 6–8 we realize it by an element of $\pi_0 \mathrm{Tot}(C^\bullet/2)$.

4.1 The ordinary isogeny cover

The coordinate S on the ordinary isogeny cover records a nonzero covector on the étale 2-torsion of the target deformation, as in Lemma 4.1. Put $U = b + S^2$. The coordinate homomorphism is

$$k[[a,b]] \longrightarrow k[[U,S]], \qquad a \longmapsto US, \quad b \longmapsto S^2 + U. \tag{4.3}$$

After inverting $a = US$, eliminating $U = a/S$ gives

$$k[[U,S]][(US)^{-1}] \cong k[[a,b]][a^{-1}][S]/(S^3 + bS + a). \tag{4.4}$$

Thus S is the universal root of the cubic, and $U = a/S = b + S^2$ is its derivative evaluated at S. Both S and U are invertible after this localization. Thus (4.4) is a finite étale algebra of rank 3. Before inverting a, the same presentation gives

$$k[[U,S]] \cong k[[a,b]][S]/(S^3 + bS + a),$$

free over $k[[a,b]]$ with basis $1, S, S^2$. Since $k[[a,b]][a^{-1}] = \pi_0(L_{K(1)}E_3)/(2)$, this is the expected shape of the cover in Proposition 2.2: for $n = 3$, $t = 1$ we have $d_{3,1} = 1 + 2 = 3$.

Lemma 4.1. *The map $\psi\colon L_{K(1)}E_3 \to L_{K(1)}(E_3^{tC_2})$ is finite étale of rank 3, and its map on degree-zero coefficients modulo 2 is the cubic extension (4.4). This identification is natural for isomorphisms of the target deformation.*

Proof. By Proposition 2.2, $\pi_0 L_{K(1)}(E_3^{tC_2})/(2)$ is the coordinate ring of the scheme of hyperplanes in the rank-two étale part of $\mathcal{G}[2]$ over $\operatorname{Spec}(k[[a,b]][a^{-1}])$. Each such hyperplane is the kernel of a unique nonzero covector over $\mathbb{F}_2$. We compute these covectors with Dieudonné theory. In the coordinates of [11, (3.3.13)], the unit-root part of the contravariant Dieudonné crystal of $\mathcal{G}$ over $k[[a,b]][a^{-1}]$ has, modulo 2, Frobenius matrix

$$\overline{F}_{\mathrm{ur}} = \begin{pmatrix} 0 & 1 \\ a^2 & b \end{pmatrix}.$$

A vector $m = (T^2/a^2,\, T/a)^{\mathsf{T}}$ satisfies $\overline{F}_{\mathrm{ur}}\,\varphi(m) = m$, where φ is the absolute Frobenius of the characteristic-2 coefficient algebra, exactly when $T^4 + bT^2 + aT = 0$, and every Frobenius-fixed vector has this form. The fixed vectors are the dual étale 2-torsion, so its nonzero elements are the roots of $T^3 + bT + a$, and their kernels are the lines we want. This gives (4.4) with $S = T$. Naturality holds because the construction only uses $\mathcal{G}[2]$, which is functorial in the target. □

Let N_0 be the kernel of the trace $k[[U,S]] \to k[[a,b]]$, and let $N = N_0[a^{-1}]$. Since $\mathrm{Tr}(1) = 3 = 1$ in k, the trace splits the unit of the cubic extension, and

$$\mathrm{pr}_N\colon k[[U,S]][(US)^{-1}] \longrightarrow N, \qquad \mathrm{pr}_N(d) = d + \mathrm{Tr}(d), \tag{4.5}$$

identifies the quotient of the cubic algebra by its unit summand with N. Since $\mathrm{Tr}(d^2) = \mathrm{Tr}(d)^2$, the projection pr_N commutes with squaring, and it is $k[[a,b]][a^{-1}]$-linear.

4.2 Coefficients of the descent diagram

We retain the solid coefficient structure through chromatic localization, as in the treatment of localized Lubin–Tate theories in [1, Section 1.3 and Proposition 4.5.10]. In the present case the required representations can be constructed from the integral lattices.

Lemma 4.2. *Let B be a finite $k[[a,b]]$-algebra with an action of $\mathbb{G}_3$ by ring automorphisms extending its action on $k[[a,b]]$. Then this action is continuous for the (a,b)-adic topology on B.*

Proof. Put $\mathcal{O}_D = \mathrm{End}_k(\Gamma_3)$, so that $\mathcal{O}_D$ is finite free over $\mathbb{Z}_2$ and $\mathbb{S}_3 = \mathcal{O}_D^\times$ is open in $\mathbb{G}_3$. On $1+4\mathcal{O}_D$ the 2-adic logarithm and exponential are mutually inverse. For every positive integer e, each $x \in 1+2^{v_2(e)+2}\mathcal{O}_D$ is an e-th power in $1+4\mathcal{O}_D$: take $\exp(e^{-1}\log x)$. Consequently any homomorphism from $\mathbb{G}_3$ to a finite group has open kernel, by taking e to be the exponent of its image.

The action preserves $(a,b)^r B$, and $B/(a,b)^r B$ is finite for every r, since $k = \mathbb{F}_8$ and B is finite over $k[[a,b]]$. Its action on each of these quotients is therefore continuous. The ring B is the inverse limit of these finite quotients, which proves joint continuity of $\mathbb{G}_3 \times B \to B$. □

The integral lattices $k[[a,b]]$, $k[[U,S]]$ and N_0 carry their adic topologies. For $k[[U,S]]$, this is also its topology as a finite $k[[a,b]]$-module; N_0 has the induced topology. These lattices are preserved by $\mathbb{G}_3$: $k[[U,S]]$ is the normalization of $k[[a,b]]$ in the cubic algebra (4.4), and the trace is equivariant. Lemma 4.2 makes the action on the finite algebra $k[[U,S]]$ continuous; the restriction to N_0 is continuous for its induced topology. A change of coordinate on the target formal group multiplies the degree-zero coefficient a of its height-one Hasse invariant by a unit of $k[[a,b]]$. They therefore define solid representations [2, Lemmas 3.2.1 and 3.3.1]. Their solid localizations are

$$\begin{aligned} \underline{k[[a,b]]}[a^{-1}] &= \varinjlim_{r\geq 0} \underline{a^{-r}k[[a,b]]}, \\ \underline{k[[U,S]]}[(US)^{-1}] &= \varinjlim_{r\geq 0} \underline{a^{-r}k[[U,S]]}, \qquad \text{in } \mathrm{Solid}^{\mathbb{G}_3}. \\ \underline{N} &= \varinjlim_{r\geq 0} \underline{a^{-r}N_0} \end{aligned}$$

Each fractional lattice has the topology transported from its integral lattice and the action inherited from its localization. For example, $g(a^{-r}m) = a^{-r}(a/g(a))^r g(m)$, so the inclusions between these lattices are equivariant. Multiplication by a is a homeomorphism from $k[[a,b]]$ onto its ideal (a) with the induced adic topology, by Artin–Rees. Hence $g \mapsto g(a)/a$ is continuous, and the displayed formula gives a continuous action on each fractional lattice. Evaluation at a point gives the ordinary localized rings in (4.4) and the trace-zero module N.

Lemma 4.3. *For $M_0 = k[[a,b]], k[[U,S]], N_0$ and $\underline{M} = \varinjlim_r \underline{a^{-r}M_0}$, the solid continuous cochains satisfy*

$$\mathrm{C}^q_{\mathrm{cts}}(\mathbb{G}_3, \underline{M}) = \mathrm{Map}_{\mathrm{cts}}(\mathbb{G}_3^q, M_0)[a^{-1}] = \varinjlim_r \mathrm{Map}_{\mathrm{cts}}(\mathbb{G}_3^q, a^{-r}M_0). \tag{4.6}$$

This identification computes $(\underline{M}^{h\mathbb{G}_3})(*)$ by the ordinary bar differential. It is compatible with the trace, the trace-zero projection and the cup products induced by continuous coefficient pairings. Replacing a by a unit multiple does not change the localized solid representation or its cohomology. The same holds if M_0 is replaced by a stable finite $k[[a,b]]$-submodule of $M_0[a^{-1}]$, with its adic topology, lying between $a^r M_0$ and $a^{-s}M_0$ for some $r,s \geq 0$.

Proof. For every profinite set T, the free solid group $\mathbb{Z}[T]^{\blacksquare}$ is compact projective [2, Lemma 3.3.2, with the trivial group]. Hence

$$\begin{aligned}\underline{M}(T) &= \mathrm{Hom}_{\mathrm{Solid}}(\mathbb{Z}[T]^{\blacksquare}, \underline{M}) \\ &= \varinjlim_{r} \mathrm{Hom}_{\mathrm{Solid}}(\mathbb{Z}[T]^{\blacksquare}, \underline{a^{-r}M_0}) \\ &= \varinjlim_{r} \mathrm{Map}_{\mathrm{cts}}(T, a^{-r}M_0).\end{aligned}$$

Taking $T = \mathbb{G}_3^q$ gives (4.6). The solidified homogeneous bar resolution is a projective resolution of the trivial representation [2, (3.3.5) and Lemma 3.3.4]. Applying equivariant Hom and then evaluating at a point therefore computes $(\underline{M}^{h\mathbb{G}_3})(*)$ by these terms. The inherited actions identify its differential with the usual inhomogeneous differential. Compact projectivity is what makes each cochain factor through one fractional lattice.

The trace and projection are continuous on the integral finite modules. Multiplication of fractional lattices adds their pole orders, so the continuous coefficient pairings also pass to solid localization. The bar construction consequently respects these maps and their cup products. A unit multiple of a gives the same fractional lattices with the same topologies. The two lattices in the last assertion give cofinal systems of fractional lattices. The inclusions are continuous for their adic topologies, by Artin–Rees, so the cofinality holds in $\mathrm{Solid}^{\mathbb{G}_3}$ and proves the last assertion. □

Proposition 4.4. *For $n = 3$ and $t = 1$, the base-change equivalence (2.1) identifies the target of (4.1) with the base change of its source along $\psi\colon L_{K(1)}E_3 \to L_{K(1)}(E_3^{tC_2})$. Both terms and their cofiber C^q have even, 2-periodic, 2-torsion-free homotopy. Their degree-zero coefficients satisfy*

$$\begin{aligned}\pi_0\big(L_{K(1)}L_{K(3)}(E_3^{\wedge(\bullet+1)})\big)/(2) &= \mathrm{C}^{\bullet}_{\mathrm{cts}}(\mathbb{G}_3, \underline{k[[a,b]]}[a^{-1}]), \\ \pi_0\big(L_{K(1)}\big((L_{K(3)}(E_3^{\wedge(\bullet+1)}))^{tC_2}\big)\big)/(2) &= \mathrm{C}^{\bullet}_{\mathrm{cts}}(\mathbb{G}_3, \underline{k[[U,S]]}[(US)^{-1}]), \\ \pi_0(C^{\bullet}/2) &= \mathrm{C}^{\bullet}_{\mathrm{cts}}(\mathbb{G}_3, \underline{N}).\end{aligned}$$

as cosimplicial modules. Under these identifications, the map from source to target is induced by $a \mapsto US$, $b \mapsto S^2 + U$, and the map from target to cofiber is induced by pr_N. Moreover $\pi_1(C^q/2) = 0$ and $\pi_2(C^{\bullet}/2) = \mathrm{C}^{\bullet}_{\mathrm{cts}}(\mathbb{G}_3, v_1\underline{N})$.

Proof. The source. The degree-q term of the completed Morava resolution satisfies

$$\pi_* L_{K(3)}(E_3^{\wedge(q+1)}) = \mathrm{Map}_{\mathrm{cts}}(\mathbb{G}_3^q, \pi_* E_3).$$

Its $K(1)$-localization, as a complex-oriented module, is inversion of $a = u_1$ followed by 2-completion [4, Proposition 3.7]. The function modules and their a-localizations are even and 2-torsion free. Their reductions modulo 2^j have surjective transition maps, so derived 2-completion has no odd homotopy and retains 2-torsion freeness. Reduction of continuous functions modulo 2 commutes with the function module: coefficientwise Witt lifts give continuous lifts from $k[[a,b]]$ to $W(k)[[a,b]]$. Localization and reduction then give

$$\begin{aligned}\pi_0\big(L_{K(1)}L_{K(3)}(E_3^{\wedge(q+1)})\big)/(2) &= \mathrm{Map}_{\mathrm{cts}}(\mathbb{G}_3^q, k[[a,b]])[a^{-1}] \\ &= \mathrm{C}^q_{\mathrm{cts}}(\mathbb{G}_3, \underline{k[[a,b]]}[a^{-1}]),\end{aligned}$$

where the last equality is Lemma 4.3.

The target. The base-change equivalence is (2.1). On coefficients modulo 2, Lemma 4.1 identifies the degree-zero target coefficients modulo 2 with $k[[U,S]][(US)^{-1}]$, and the Tate Frobenius sends a function f to $g \mapsto \psi(f(g))$. The basis $1, S, S^2$ identifies $k[[U,S]]$ with three copies of $k[[a,b]]$ as a topological module. Continuous functions therefore commute with this finite base change; inverting a gives the asserted target coefficient identity. The structure maps are those of Proposition 2.2.

The cofiber. The trace of the finite étale extension in (4.1) splits its unit, since its value on 1 is 3, a unit. Thus C^q is the complementary summand, finite projective over the source term; its homotopy is even, periodic and torsion free, and by (4.5) its mod-2 coefficients are $\mathrm{C}^{\bullet}_{\mathrm{cts}}(\mathbb{G}_3, \underline{N})$. Thus the coefficient diagrams are the bar diagrams of the localized solid representations. As the homotopy of C^q is even and torsion free, $\pi_1(C^q/2) = 0$.

Degree two. It remains to identify the generator of π_2 modulo 2 in a $\mathbb{G}_3$-equivariant way. Over $k[[a,b]][a^{-1}]$, the formal group $\mathcal{G}$ modulo 2 has height one, and an isomorphism from it to $\widehat{\mathbb{G}}_m$ has derivative w at the origin with $wa = w^2$, since a is the coefficient of x^2 in $[2](x)$ and $[2](x) = x^2$ on $\widehat{\mathbb{G}}_m$ modulo 2. As w is a unit, $w = a$. Pulling back the invariant differential, the Bott class of KU corresponds to $au^{-1} = v_1$, and this is natural for isomorphisms of formal groups, hence $\mathbb{G}_3$-equivariant. □

5 An arithmetic class at height three

Let $\Lambda = T_2(\mathcal{G}^{\mathrm{et}})$ be the rank-two étale $\mathbb{Z}_2$-local system over $\mathrm{Spec}(k[[a,b]][a^{-1}])$ given by the Tate module of the étale quotient of $\mathcal{G}$. By Lemma 4.1, the cover $\mathrm{Spec}(k[[U,S]][(US)^{-1}]) \to \mathrm{Spec}(k[[a,b]][a^{-1}])$ parametrizes one-dimensional $\mathbb{F}_2$-subspaces of $\Lambda/2$. Write ℓ for its universal line. A lift of ℓ is a rank-one $\mathbb{Z}/4$-direct summand of the pullback of $\Lambda/4$ whose reduction modulo 2 is ℓ. Let $Z \to \mathrm{Spec}(k[[U,S]][(US)^{-1}])$ be the scheme parametrizing these lifts. It is finite étale of degree 2; interchanging the two lifts in each geometric fiber gives its canonical $\mathbb{Z}/2$-torsor structure. The action of $\mathbb{G}_3$ on Λ induces an action on Z commuting with its deck involution. We will show that this double cover gives a nonzero class in $H^1(\mathbb{G}_3, \underline{N})(*)$.

In a frame for $\Lambda/4$ in which ℓ is the first coordinate line modulo 2, the subgroup of $\mathrm{GL}_2(\mathbb{Z}/4)$ preserving ℓ is

$$P = \left\{ \begin{pmatrix} r & s \\ 2w & r' \end{pmatrix} : \ r, r' \text{ odd} \right\}.$$

The two lifts of ℓ are spanned by $(1,0)$ and $(1,2)$, and P permutes them through the character

$$x_P \colon P \longrightarrow \mathbb{F}_2, \qquad x_P \begin{pmatrix} r & s \\ 2w & r' \end{pmatrix} = w \bmod 2. \tag{5.1}$$

Thus Z is associated to the P-torsor of adapted frames via the character x_P.

Proposition 5.1. *There is a regular function $z \in \Gamma(Z, \mathcal{O}_Z)$ with $z \mapsto z+1$ under the deck involution such that, for every $g \in \mathbb{G}_3$, the function $gz - z$ descends to $\mathrm{Spec}(k[[U,S]][(US)^{-1}])$. The resulting maps*

$$x_{\mathrm{line}} \colon \mathbb{G}_3 \longrightarrow a^{-2}k[[U,S]], \quad g \longmapsto gz - z,$$
$$x_N \colon \mathbb{G}_3 \longrightarrow a^{-2}N_0, \qquad g \longmapsto \mathrm{pr}_N(x_{\mathrm{line}}(g))$$

are continuous 1-cocycles for the adic topologies on these fractional lattices. With the localized solid coefficient structures, they define classes with coefficients in $\underline{k[[U,S]][(US)^{-1}]}$ and $\underline{N}$, respectively,

and

$$[x_N] \in H^1(\mathbb{G}_3, \underline{N})(*)$$

is nonzero. For the central element $c = [-1] \in \mathbb{G}_3$, $x_{\mathrm{line}}(c) = x_N(c) = 0$.

We first compute an Artin–Schreier equation for the double cover (Lemma 5.2) and prove that its trace-zero parameter is not an Artin–Schreier boundary (Lemma 5.3). The vanishing of the $\mathbb{S}_3$-invariants of N (Proposition 5.4) then excludes a coboundary for x_N, as shown at the end of the section.

5.1 Lifts of lines modulo four

Lemma 5.2. *Let* $Z \to \mathrm{Spec}(k[[U,S]][(US)^{-1}])$ *be the* $\mathbb{Z}/2$*-torsor of lifts of the universal line* $\ell \subset \Lambda/2$ *to rank-one* $\mathbb{Z}/4$*-direct summands of* $\Lambda/4$*. Its coordinate ring has the following presentation as an algebra over* $k[[U,S]][(US)^{-1}]$*:*

$$\begin{aligned} \Gamma(Z, \mathcal{O}_Z) &\cong k[[U,S]][(US)^{-1}][z]/(z^2 + z - d), \\ d &= \frac{S^6}{U^3} + \frac{S}{U^4} = \frac{S^9}{a^3} + \frac{S^5}{a^4}. \end{aligned} \tag{5.2}$$

Under this isomorphism the nontrivial deck transformation sends z *to* $z+1$*. The* $\mathbb{G}_3$*-action on this algebra is the action transported from the moduli of line lifts.*

Proof. The idea is to compute the lifts of ℓ on the Dieudonné side. The dual of the étale Tate module modulo 4 is realized as the étale sheaf of Frobenius-fixed vectors of an explicit unit-root crystal. A lift of ℓ is described by a Frobenius-stable line in the dual unit-root module, and Frobenius stability gives a quadratic equation modulo 2 for its correction term.

The unit-root crystal modulo 4. All computations in this proof are modulo 4, over the lift $W_2(k)[[a,b]][a^{-1}]$ of $k[[a,b]][a^{-1}]$, on which the Frobenius lift φ is the Witt vector Frobenius on constants, $\varphi(a) = a^2$ and $\varphi(b) = b^2$. Let $\mathscr{H}$ be the contravariant Dieudonné crystal of $\mathcal{G}$, with its connection and φ-semilinear Frobenius Φ. In a basis E_0, E_1, E_2, the matrix of Φ is [11, (3.3.13) and Example 3.3.18]

$$\begin{pmatrix} 0 & 0 & 1 \\ 2 & 0 & -a \\ 0 & 1 & -b \end{pmatrix}.$$

The vectors

$$g_0 = E_0 + \Big(-a + \frac{2b}{a^2}\Big)E_1, \qquad g_1 = E_2 + \frac{2}{a^2}E_1$$

satisfy $\Phi(g_0) = (-a^2 + 2b^2/a^4)\, g_1$ and $\Phi(g_1) = g_0 + (-b + 2/a^4)\, g_1$, so they span a Frobenius-stable submodule M with matrix

$$F_{\mathrm{ur}} = \begin{pmatrix} 0 & 1 \\ -a^2 + 2b^2/a^4 & -b + 2/a^4 \end{pmatrix}. \tag{5.3}$$

Its determinant is a unit. The vectors g_0, g_1, E_1 form a basis, so M is a rank-two direct summand of the underlying module $\mathscr{H}$; the quotient $\mathscr{H}/M$ has Frobenius $-2/a^2$. We identify M with the étale subcrystal. Write $\mathbb{D}$ for the contravariant crystalline Dieudonné functor. Its level-two truncation satisfies $\mathbb{D}(H[4]) = \mathbb{D}(H)/4$ for a 2-divisible group H. Apply its exactness

to the connected–étale sequence on 4-torsion and evaluate on the divided-power thickening $\operatorname{Spec}(k[[a,b]][a^{-1}]) \hookrightarrow \operatorname{Spec}(W_2(k)[[a,b]][a^{-1}])$. This gives

$$0 \longrightarrow \mathbb{D}(\mathcal{G}^{\mathrm{et}})/4 \longrightarrow \mathscr{H} \longrightarrow \mathbb{D}(\mathcal{G}^{0})/4 \longrightarrow 0,$$

where all crystals in this sequence denote their evaluations. The end terms are locally free of ranks 2 and 1. On the étale term the linearized Frobenius is invertible. We use here the truncated exactness and the constant-group calculation recalled in [8, §4.3, pp. 41–42].

Let $f\colon \mathbb{D}(\mathcal{G}^{\mathrm{et}})/4 \to \mathscr{H}/M$ be the composite, and write Φ_{et} and $\Phi_{\mathscr{H}/M}$ for the two linearized Frobenius maps. For either map put $\Phi^{(2)} = \Phi \circ \varphi^{*}\Phi$. Frobenius compatibility gives

$$f \circ \Phi_{\mathrm{et}}^{(2)} = \Phi_{\mathscr{H}/M}^{(2)} \circ \varphi^{2*} f = 0,$$

since the quotient Frobenius is divisible by 2 and $4 = 0$. Since $\Phi_{\mathrm{et}}^{(2)}$ is invertible, $f = 0$. Thus $\mathbb{D}(\mathcal{G}^{\mathrm{et}})/4 \subset M$. The quotient of $\mathscr{H}$ by this submodule is locally free, so its inclusion remains injective on every residue-field fiber. It factors through the rank-two fiber of M and is therefore an isomorphism there. Nakayama's lemma gives $M = \mathbb{D}(\mathcal{G}^{\mathrm{et}})/4$. In particular M is a subcrystal and is preserved by the connection.

The comparison with the Tate module is given by evaluation. An étale-local section t of $\Lambda/4 = \mathcal{G}^{\mathrm{et}}[4]$ defines $t\colon \mathbb{Z}/4 \to \mathcal{G}^{\mathrm{et}}[4]$. Contravariance and $\mathbb{D}(\mathbb{Z}/4) = \mathcal{O}_{\mathrm{crys}}/4$ give a pairing of crystals

$$(\mathbb{D}(\mathcal{G}^{\mathrm{et}})/4) \otimes_{\mathbb{Z}/4} (\Lambda/4) \longrightarrow \mathcal{O}_{\mathrm{crys}}/4.$$

It is perfect, as is seen after an étale trivialization of $\mathcal{G}^{\mathrm{et}}[4]$. On the constant factor the connection is trivial and Frobenius acts by the coefficient Frobenius. Katz's correspondence [14, Proposition 4.1.1] applies: the chosen lift is flat over $W_2(k)$ and its special fiber $k[[a,b]][a^{-1}]$ is normal and integral. It identifies the étale sheaf of fixed sections as

$$M^{\Phi=1} \simeq \underline{\mathrm{Hom}}_{\mathbb{Z}/4}(\Lambda/4, \mathbb{Z}/4) = (\Lambda/4)^{\vee}.$$

The pairing and the inclusion of the étale subcrystal are natural under base change and isomorphisms of the target group. This makes the identification $\mathbb{G}_3$-equivariant. For semilinear coordinate changes, this naturality uses crystalline pullback and the connection's comparison of Frobenius lifts.

The line modulo 2. Modulo 2, F_{ur} is the matrix $\overline{F}_{\mathrm{ur}}$ of Lemma 4.1, its fixed vectors are $m_T = (T^2/a^2, T/a)^{\mathsf{T}}$ with $T^3 + bT + a = 0$, and the root $T = S$ gives the linear form m_S on the Tate module whose kernel is ℓ. Dually, ℓ corresponds to the row vector $(-U, 1)$ in the dual of M, since $(-U, 1)\, m_S = -US^2/a^2 + S/a = 0$ by $a = US$.

The lifts. Over the unique étale lift of $k[[U,S]][(US)^{-1}]$ modulo 4, lift U, S so that $a = US$ and $b = S^2 + U$. The Frobenius lift is

$$\varphi(U) = U^2 + 2US^2, \qquad \varphi(S) = S^2 + \frac{2S^4}{U}. \tag{5.4}$$

These satisfy $\varphi(a) = a^2$ and $\varphi(b) = b^2$, and the Frobenius lift of an étale algebra is unique. A lift of ℓ is spanned by a row $r = (-U + 2v, 1)$. It is Frobenius-stable when $r\, F_{\mathrm{ur}} = \lambda\, \varphi(r)$, where λ is the Frobenius multiplier relative to the chosen row r. The second coordinate determines this multiplier: $\lambda = -S^2 - 2U + 2v + 2/a^4$. With this λ, the first coordinates agree modulo 2, and comparing the coefficients of 2 gives

$$S^2v^2 + U^2v = U^2S^2 + U^3 + US^4 + U^{-4} \pmod 2. \tag{5.5}$$

Put $z_0 = S^2 v/U^2$. Multiplying (5.5) by S^2/U^4 gives

$$z_0^2 + z_0 = \frac{S^4}{U^2} + \frac{S^2}{U} + \frac{S^6}{U^3} + \frac{S^2}{U^8},$$

and $z = z_0 + S^2/U + S/U^4$ satisfies $z^2 + z = d$ with d as in (5.2). The two solutions v and $v + U^2/S^2$ correspond to z and $z + 1$.

Horizontality. It remains to see that the Frobenius-stable lines found above are the lifts of ℓ in $\Lambda/4$, that is, that they are horizontal. The row line spanned by r lies in the dual of M; it is horizontal if and only if its annihilator $L = \ker r \subset M$ is horizontal, where r is viewed as a linear functional on M. Let $\kappa\colon L \to (M/L) \otimes \Omega^1$ be its second fundamental form. It is compatible with Frobenius. Since $\lambda \equiv S^2$ modulo 2, the linearized Frobenius is invertible on L and on M/L, whereas $\mathrm{d}\varphi$ is divisible by 2. So the Frobenius relation writes κ as an invertible conjugate of $(1 \otimes \mathrm{d}\varphi)\varphi^*\kappa$, which is divisible by 2; iterating once more, κ is divisible by 4, hence $\kappa = 0$. All changes of variables above are reversible, since U and S are units. They therefore identify horizontal line lifts with solutions of $z^2 + z = d$ after every étale base change. This proves the asserted coordinate-ring presentation, with the deck involution $z \mapsto z + 1$. □

Lemma 5.3. *Let d be the parameter in (5.2), and put $d_N = \mathrm{pr}_N(d) \in N$. Then*

$$\mathrm{Tr}(d) = 1 + \frac{b^3}{a^2} + \frac{b}{a^3}, \qquad d_N = d + 1 + \frac{b^3}{a^2} + \frac{b}{a^3}. \tag{5.6}$$

Under the embedding $k[[U,S]][(US)^{-1}] \hookrightarrow k((S))((U))$ obtained by expansion in U, the equation $h^2 + h = d_N$ has no solution in $k((S))((U))$. In particular it has no solution in N.

Proof. From $S^3 = bS + a$ in characteristic 2 one gets the power sums $\mathrm{Tr}(S^5) = ab$ and $\mathrm{Tr}(S^9) = ab^3 + a^3$; substituting into the second expression for d in (5.2) gives $\mathrm{Tr}(d)$, and then d_N by (4.5).

The ring $k[[U,S]][(US)^{-1}]$ embeds in the field $k((S))((U))$. Substituting $a = US$ and $b = S^2 + U$,

$$d_N = \frac{S}{U^4} + \frac{S^6 + S^{-1}}{U^3} + \frac{S^4 + S^{-3}}{U^2} + \frac{S^2}{U} + \frac{U}{S^2}. \tag{5.7}$$

Suppose $h^2 + h = d_N$. Since d_N has a pole in U, so has h; if the pole has order $m > 0$, then $h^2 + h$ has a pole of order $2m$ with leading coefficient the square of that of h. Comparing with the term S/U^4 gives $m = 2$ and a leading coefficient $c \in k((S))$ with $c^2 = S$. This is impossible, since S has odd valuation. □

5.2 Stabilizer invariants

Proposition 5.4. *For the actions induced by the universal deformation and its ordinary isogeny cover, the pure stabilizer group $\mathbb{S}_3$ has*

$$(k[[a,b]][a^{-1}])^{\mathbb{S}_3} = k, \qquad (k[[U,S]][(US)^{-1}])^{\mathbb{S}_3} = k, \qquad N^{\mathbb{S}_3} = 0.$$

The elements of $k[[a,b]][a^{-1}]$ are functions on the height-one stratum of the Lubin–Tate space in characteristic 2. The two-tower isomorphism presents this stratum, at infinite level, as a torsor over a period domain, with group $\mathbb{S}_3$. An invariant function therefore descends to the period domain, and the main step is that the period domain has only constant tilted functions. On its connected Drinfeld cover, compatible roots of all 2-power orders force the derivative of such a function to vanish.

Proof. It suffices to prove $(k[[a,b]][a^{-1}])^{\mathbb{S}_3} = k$. Indeed, let $f \in (k[[U,S]][(US)^{-1}])^{\mathbb{S}_3}$. The coefficients of its characteristic polynomial over $k[[a,b]][a^{-1}]$ are invariant, hence in k, so f is algebraic over k. Since $f \in \mathrm{Frac}(k[[U,S]])$ is integral over k, normality gives $f \in k[[U,S]]$ and $f^{8^m} = f$ for some m. By Hensel's lemma (the derivative of $X^{8^m} - X$ is -1), f equals its residue in k. Finally $N^{\mathbb{S}_3} = N \cap k = 0$, because $\mathrm{Tr}(1) = 1 \neq 0$.

Fix a completed algebraic closure $\mathbb{C}_2$ of $W(k)[1/2]$ and put $\Delta = \mathrm{Gal}(\overline{\mathbb{Q}}_2/W(k)[1/2])$. Let $\mathrm{LT}_{3,1}$ be the height-one stratum of the Lubin–Tate space of Γ_3 in characteristic 2, so that $k[[a,b]][a^{-1}]$ is the ring of functions on it with a finite pole along $a = 0$, and let $\mathrm{LT}_{3,1}^{\infty}$ be its infinite-level cover. By [1, Proposition 2.6.6(2) and Lemma 2.4.3] there are torsors of v-sheaves over k

$$\mathrm{LT}_{3,1}^{\diamond} \longleftarrow \mathrm{LT}_{3,1}^{\infty} \longrightarrow \Omega_{3,1}, \tag{5.8}$$

the first pro-finite étale with group $\mathrm{Aut}(\Gamma_1) \times \mathrm{GL}_2(\mathbb{Z}_2)$ and the second with group $\mathbb{S}_3$, commuting with each other. Here $\Omega_{3,1}$ classifies extensions $0 \to \mathcal{O}^2 \to \mathcal{E} \to \mathcal{O}(1) \to 0$ on the relative Fargues–Fontaine curve with $\mathcal{E}$ isoclinic of slope $1/3$.

We prove three facts:

$$\mathbb{C}_2^{\flat} \xrightarrow{\sim} H^0(\Omega_{3,1,\mathbb{C}_2}, \mathcal{O}^{\flat}), \tag{5.9}$$

$$k[[a,b]][a^{-1}] \hookrightarrow H^0(\mathrm{LT}_{3,1,\mathbb{C}_2}^{\infty}, \mathcal{O}^{\flat}), \tag{5.10}$$

$$(\mathbb{C}_2^{\flat})^{\Delta} = k. \tag{5.11}$$

Given these, let $f \in (k[[a,b]][a^{-1}])^{\mathbb{S}_3}$. Its pullback to $\mathrm{LT}_{3,1,\mathbb{C}_2}^{\infty}$ is $\mathbb{S}_3$-invariant, so it descends to $\Omega_{3,1,\mathbb{C}_2}$ and is a constant in $\mathbb{C}_2^{\flat}$ by (5.9). It is also fixed by Δ, which acts through the base change and fixes functions coming from $k[[a,b]][a^{-1}]$, so the constant lies in k by (5.11). By (5.10), f equals this constant.

Proof of (5.9). By [1, Proposition 3.1.2(1)] the Banach–Colmez space $\mathrm{BC}(\mathcal{O}(-1))$ over $\mathbb{C}_2$ is $\mathbb{A}^1_{\mathbb{C}_2}/\mathbb{Q}_2$, and the extension class of $\mathcal{E}$ is given by two coordinates in it. On each geometric fiber, $\mathcal{E}$ has degree 1 and slopes in $[0,1]$, so it is $\mathcal{O}(1/s) \oplus \mathcal{O}^{3-s}$ for some s, and it is isoclinic of slope $1/3$ exactly when $\mathrm{Hom}(\mathcal{E}, \mathcal{O}) = 0$. Applying $\mathrm{Hom}(-, \mathcal{O})$ to the extension, this means that the map $\mathbb{Q}_2^2 \to \mathrm{BC}(\mathcal{O}(-1))$ given by the two coordinates is injective. Hence the inverse image of $\Omega_{3,1,\mathbb{C}_2}$ in $\mathbb{A}^2_{\mathbb{C}_2}$ is Drinfeld's symmetric space, written in the affine chart with first homogeneous coordinate 1:

$$H_{\mathbb{C}_2} = \{(z_1, z_2) : 1, z_1, z_2 \text{ linearly independent over } \mathbb{Q}_2\}.$$

This space is smooth and geometrically connected [9, Theorem 1.1], and maps onto $\Omega_{3,1,\mathbb{C}_2}$ by a quotient v-cover. By descent it suffices to show that $H_{\mathbb{C}_2}$ has only constant tilted functions. Let $f \in H^0(H_{\mathbb{C}_2}, \mathcal{O}^{\flat})$ and $f_j = (\mathrm{Frob}^{-j} f)^{\sharp}$; these are analytic functions [10, Proposition 2.25] with $f_j^{2^j} = f_0$. On a closed polydisc of radius ρ, the derivative of $f_0 = f_j^{2^j}$ is divisible by 2^j, so

$$\|\mathrm{d}f_0\| \leq |2|^j \rho^{-1} \|f_j\|^{2^j} = |2|^j \rho^{-1} \|f_0\|.$$

Letting $j \to \infty$ gives $\mathrm{d}f_0 = 0$, so f_0 is locally constant, hence constant because $H_{\mathbb{C}_2}$ is connected. The same holds for every f_j, and the compatible system (f_j) is an element of $\mathbb{C}_2^{\flat}$.

Proof of (5.10). Let $0 \neq f \in k[[a,b]][a^{-1}]$. Choose $\varpi \in \mathbb{C}_2^{\flat}$ with $0 < |\varpi| < 1$. The substitution $a \mapsto \varpi$, $b \mapsto \varpi T$ embeds $k[[a,b]][a^{-1}]$ into the Tate algebra $k((\varpi))\langle T \rangle$: a series $\sum_i h_i(a) b^i$ in $k[[a,b]]$ maps to $\sum_i h_i(\varpi) \varpi^i T^i$, whose coefficients tend to zero and vanish only if $h_i = 0$. By the Nullstellensatz for affinoid algebras [7, Theorem 1.1.5], there is a maximal ideal at which f does not vanish, with residue field finite over $k((\varpi))$. It embeds into $\mathbb{C}_2^{\flat}$ compatibly with valuations,

which gives a point of $\mathrm{LT}_{3,1}$ with $0 < |a| < 1$, $|b| < 1$ and $f \neq 0$. Over an algebraically closed perfectoid field the fibers of the first torsor in (5.8) are nonempty profinite sets, so the point lifts to infinite level, where the pullback of f is still nonzero.

Proof of (5.11). By Tate–Sen, $\mathbb{C}_2^{\Delta} = W(k)[1/2]$ [6, Theorem 2.2.7]. A nonzero Δ-fixed element of $\mathbb{C}_2^{\flat}$ is a sequence (y_j) in $W(k)[1/2]$ with $y_{j+1}^2 = y_j$. Since the valuation is discrete, every y_j is a unit. Dividing by the Teichmüller lifts of the residues, we may assume $y_j \in 1 + 2W(k)$ for all j. Then $y_0 = y_j^{2^j} \in 1 + 2^{j+1}W(k)$ for every j, so $y_0 = 1$, and likewise every $y_j = 1$. Hence the element is the Teichmüller sequence of an element of k. □

5.3 The class

Proof of Proposition 5.1. Let z be the coordinate of Lemma 5.2. The action of $\mathbb{G}_3$ on Z commutes with the deck involution $z \mapsto z + 1$, so $x_{\mathrm{line}}(g) = gz - z$ is invariant under it and is a function on $\mathrm{Spec}(k[[U,S]][(US)^{-1}])$. It satisfies $x_{\mathrm{line}}(gh) = x_{\mathrm{line}}(g) + g\, x_{\mathrm{line}}(h)$, so it is a cocycle representing the descent class of the double cover attached to x_P. Replacing z by $z+h$ for $h \in k[[U,S]][(US)^{-1}]$ adds the coboundary $g \mapsto gh - h$ to x_{line}. Thus its projection defines a coordinate-independent class $[x_N]$.

Pole bound and continuity. The group $\mathbb{G}_3$ preserves $k[[U,S]]$, the normalization of $k[[a,b]]$ in the cubic algebra, and multiplies a by units of $k[[a,b]]$. From $z^2 + z = d$ we get

$$x_{\mathrm{line}}(g)^2 + x_{\mathrm{line}}(g) = g\,d - d \in a^{-4}k[[U,S]].$$

Along a height-one prime of $k[[U,S]]$ where $x_{\mathrm{line}}(g)$ has a pole, the left side has a pole of twice the order. The divisor of $a = US$ is reduced, so $x_{\mathrm{line}}(g)$ has a pole of order at most 2 along $U = 0$ and along $S = 0$, and no other pole. As $k[[U,S]]$ is normal, $x_{\mathrm{line}}(g) \in a^{-2}k[[U,S]]$ for every g, and then $x_N(g) \in a^{-2}N_0$.

For continuity, take the normalization of $k[[U,S]]$ in the finite étale algebra of Z. It is finite over the excellent ring $k[[U,S]]$, hence also over $k[[a,b]]$. The natural action on Z preserves this normalization and extends the action on $k[[a,b]]$. It is continuous by Lemma 4.2; the (a,b)-adic and (U,S)-adic topologies agree on this finite module. Moreover, $a^2 z$ is integral, since it satisfies

$$(a^2 z)^2 + a^2(a^2 z) = a^4 d \in k[[U,S]].$$

Consequently the orbit map of $a^2 z$ is continuous in this normalization, and

$$a^2 x_{\mathrm{line}}(g) = \frac{a^2}{g(a)^2}\, g(a^2 z) - a^2 z$$

is continuous there. Its values lie in $k[[U,S]]$, whose induced topology is its adic topology by Artin–Rees. Thus x_{line} is continuous in the single lattice $a^{-2}k[[U,S]]$. The finite-module trace projection is continuous and preserves the lattice, which proves the asserted continuity of x_N. By Lemma 4.3, these maps are scalar continuous cochains with coefficients in $\underline{k[[U,S]][(US)^{-1}]}$ and $\underline{N}$, respectively.

Nonvanishing. The projection pr_N is $\mathbb{G}_3$-equivariant and commutes with squaring, so applying it to the Artin–Schreier identity above gives

$$g\,d_N - d_N = x_N(g)^2 + x_N(g) \qquad (g \in \mathbb{G}_3).$$

A zero-cochain with coefficients in $\underline{N}$ is an element of $\underline{N}(*) = N$. Suppose $x_N(g) = g\nu - \nu$ for some $\nu \in N$ and all $g \in \mathbb{S}_3$. Then $g\,d_N - d_N = g(\nu^2 + \nu) - (\nu^2 + \nu)$, so $d_N + \nu^2 + \nu \in N$ is

$\mathbb{S}_3$-invariant. By Proposition 5.4 this element is zero, so $d_N = \nu^2 + \nu$, contradicting Lemma 5.3. Hence x_N is not a coboundary on $\mathbb{S}_3$, and a fortiori not on $\mathbb{G}_3$.

The central element. The element $c = [-1]$ acts trivially on the deformation space, and on Tate modules as the scalar -1, which fixes every line modulo 2 and modulo 4. So c acts trivially on Z, and $x_{\text{line}}(c) = x_N(c) = 0$. □

6 The transfer and height-one descent

We now produce an element of $\pi_0 \operatorname{Tot}(C^\bullet/2)$ from the double cover of Section 5, and set up the comparison with K-theory that will compute its first nonzero invariant.

6.1 The transfer

Proposition 6.1. *The reduced stable transfer of the double cover $Z \to \operatorname{Spec}(k[[U,S]][(US)^{-1}])$, equivalently of the P-set $P/\ker(x_P)$, means its stable transfer minus the transfer of two points. Via the level-4 frame torsor and the map to the cofiber $C^\bullet/2$, it defines an element*

$$\operatorname{tr}_C \in \pi_0 \operatorname{Tot}(C^\bullet/2)$$

whose image under every evaluation map $\pi_0 \operatorname{Tot}(C^\bullet/2) \to \pi_0(C^q/2)$ is zero.

Proof. The level-4 frames of the rank-two étale Tate module form a finite étale $\mathrm{GL}_2(\mathbb{Z}/4)$-torsor over the degree-zero mod-2 coefficients of the source term in (4.1). This term is 2-complete. Henselian lifting, followed by [16, Theorem 7.5.0.6], therefore gives a finite étale algebra T^q over it, the level-4 frame algebra. The frame torsor is functorial in the target formal group, as in Lemma 4.1, so by uniqueness of the lifts the T^q form a cosimplicial $\mathrm{GL}_2(\mathbb{Z}/4)$-torsor. Its quotient by P is the cover of lines, so

$$L_{K(1)}\Big((L_{K(3)}(E_3^{\wedge(q+1)}))^{tC_2}\Big) \simeq (T^q)^{hP}; \tag{6.1}$$

to justify this also for periodic spectra, first truncate above any integer m. The torsor identity identifies the group cochains on each $\pi_j T^q$ with its faithfully flat Amitsur complex, so its cohomology is $\pi_j L_{K(1)}\Big((L_{K(3)}(E_3^{\wedge(q+1)}))^{tC_2}\Big)$ in degree zero and zero in higher degrees. The homotopy fixed point spectral sequence for $\tau_{\le m} T^q$ has, in total degree d, only the finitely many terms with $0 \le s \le m - d$. It therefore gives a natural equivalence

$$\tau_{\le m} L_{K(1)}\Big((L_{K(3)}(E_3^{\wedge(q+1)}))^{tC_2}\Big) \xrightarrow{\sim} (\tau_{\le m} T^q)^{hP}. \tag{6.2}$$

Spectra are left complete, and homotopy fixed points commute with limits; taking the inverse limit over m proves (6.1). The same argument applies modulo 2, since $\pi_* L_{K(1)}\Big((L_{K(3)}(E_3^{\wedge(q+1)}))^{tC_2}\Big)$ and $\pi_* T^q$ are even and 2-torsion free. These equivalences are natural in the torsor, hence in q.

Let $\operatorname{tr}_P = [P/\ker x_P] - 2 \in \pi_0(\mathbb{S}^{hP})$ be the stable transfer of the P-set $P/\ker x_P$ minus that of two points. The units $\mathbb{S} \to T^q$ are compatible with the cosimplicial maps, so via (6.1) tr_P gives an element of $\pi_0 \operatorname{Tot}_q L_{K(1)}\Big((L_{K(3)}(E_3^{\wedge(q+1)}))^{tC_2}\Big)$, and we let tr_C be its image under the quotient map to $\operatorname{Tot}(C^\bullet/2)$. On T^q the P-set restricts to two points, so the image of tr_P in $\pi_0 T^q$ is $2 - 2 = 0$. The coefficient map from the Tate term to the frame algebra is faithfully flat, hence injective. Thus the transfer vanishes in the degree-zero coefficients of the Tate term and of $C^q/2$. □

Since $\pi_1(C^q/2) = 0$ and tr_C vanishes on π_0 coefficients, its first possible invariant lives in H^2 of the π_2 coefficients. We make this precise.

Lemma 6.2. *Let $M^\bullet$ be a cosimplicial spectrum with $\pi_1 M^q = 0$ for all q, and let $y \in \pi_0 \operatorname{Tot} M^\bullet$ map to zero in $H^0(\pi_0 M^\bullet)$. Here $H^i(\pi_j M^\bullet)$ denotes the cohomology of the normalized complex of the cosimplicial abelian group $q \mapsto \pi_j M^q$, and all Postnikov truncations below are applied objectwise. Then y lifts uniquely to $\pi_0 \operatorname{Tot} \tau_{\geq 2} M^\bullet$. Projecting this lift along $\operatorname{Tot} \tau_{\geq 2} M^\bullet \to \operatorname{Tot} \Sigma^2 H(\pi_2 M^\bullet)$ defines its second Postnikov-layer class, denoted by*

$$\mathrm{pr}_2(y) \in H^2(\pi_2 M^\bullet).$$

If $\mathrm{pr}_2(y) \neq 0$ then $y \neq 0$. The construction is natural in maps of cosimplicial spectra.

Proof. Totalize the fiber sequence $\tau_{\geq 2} M^\bullet \to M^\bullet \to \tau_{\leq 1} M^\bullet$. Since $\pi_1 M^q = 0$, each $\tau_{\leq 1} M^q$ is coconnective with $\pi_0 = \pi_0 M^q$, so $\operatorname{Tot} \tau_{\leq 1} M^\bullet$ has $\pi_1 = 0$ and $\pi_0 = H^0(\pi_0 M^\bullet)$. The long exact sequence shows that y lifts uniquely and that $\pi_0 \operatorname{Tot} \tau_{\geq 2} M^\bullet \to \pi_0 \operatorname{Tot} M^\bullet$ is injective. The bottom layer $\Sigma^2 H(\pi_2 M^\bullet)$ has totalization with $\pi_0 = H^2(\pi_2 M^\bullet)$. So a nonzero $\mathrm{pr}_2(y)$ shows that the lift, and hence y, is nonzero. □

The same construction applies to continuous homotopy fixed points M^{hG}, using the cosimplicial resolution by continuous cochains; then $\mathrm{pr}_2(y) \in H^2_{\mathrm{cts}}(G, \pi_2 M)$. In the present application, Proposition 4.4 and Lemma 4.3 identify

$$H^2(\pi_2(C^\bullet/2)) \cong H^2(\mathbb{G}_3, v_1 \underline{N})(*).$$

This is the comparison of the second Eilenberg–Mac Lane layers: the coefficient diagram is the scalar solid bar complex, and evaluation at a point is exact. The periodic spectra themselves continue to be compared by their Postnikov truncations.

6.2 Descent along $KU_2^\wedge$

To compute $\mathrm{pr}_2(\mathrm{tr}_C)$, we compare with the universal case, K-theory with a trivial action of C_2, where the transfer is the class of a line bundle. The comparison goes through $KU_2^\wedge$ with its continuous action of $\mathbb{Z}_2^\times$ by Adams operations. For the descent along $KU_2^\wedge \wedge (-)$ of $K(1)$-local spectra we use [5, Lemma 3.3 and Proposition 3.10]; modulo 2 the relevant coefficient modules are smooth, that is, each element is fixed by an open subgroup.

Lemma 6.3. *Let A be an even-periodic complex-oriented 2-local $\mathbb{E}_\infty$-ring with $\pi_* A$ torsion free and v_1 invertible modulo 2, and let $\mathcal{F}_A$ be its formal group over the ring $\pi_0 A/(2)$. Then $L_{K(1)}(KU_2^\wedge \wedge A)/2 \simeq (KU_2^\wedge \wedge A)/2$, and its degree-zero homotopy ring is*

$$\pi_0\big((KU_2^\wedge \wedge A)/2\big) \cong \varinjlim_{m \geq 1} \mathcal{O}\big(\mathrm{Isom}(\mathcal{F}_A[2^m], \mu_{2^m})\big). \tag{6.3}$$

Here Isom *denotes isomorphisms of finite flat group schemes over $\pi_0 A/(2)$. The transition maps are pullback along restriction of isomorphisms from level 2^{m+1} to level 2^m. Each* Isom *scheme is a finite étale $(\mathbb{Z}/2^m)^\times$-torsor over* $\mathrm{Spec}(\pi_0 A/(2))$. *This is natural in A and in finite étale base change.*

Proof. Multiplication by 2 is injective on $\pi_* A$, and multiplication by v_1 is injective on $\pi_* A/2$ because it is a unit there. The further Landweber quotients are zero. Landweber's criterion [15, 2.6] therefore applies and gives

$$\pi_*(A \wedge Y) \cong \pi_* A \otimes_{MU_*} MU_* Y \tag{6.4}$$

for every spectrum Y. The cooperation calculation is the base change of the Hopf algebroid of formal group laws and strict isomorphisms. Landweber's flatness criterion for KU says that its map to the moduli stack of one-dimensional formal groups is flat. Pulling this map back along the formal group of A shows that the cooperation algebra in (6.4), for $Y = KU$, is flat over $\pi_* A$. It is even and 2-torsion free, so its derived reduction modulo 2 is its algebraic quotient, with no odd Tor groups. This also computes $(KU_2^\wedge \wedge A)/2$, since $KU/2 \simeq KU_2^\wedge/2$. The spectrum $(KU_2^\wedge \wedge A)/2$ is a $KU_2^\wedge$-module, hence $E(1)$-local, and its homotopy is killed by 2, so it is 2-complete; for such modules $K(1)$-localization is 2-completion, which proves the first assertion.

The same Hopf algebroid calculation identifies $\pi_0((KU_2^\wedge \wedge A)/2)$ with the coordinate ring of $\mathrm{Isom}(\mathcal{F}_A, \widehat{\mathbb{G}}_m)$. For periodic theories the ratio of the periodicity generators records the derivative, so this represents all isomorphisms, rather than only strict ones. The formal group $\mathcal{F}_A$ has height one, so $[2^m](x) = x^{2^m} \cdot (\text{unit})$, and the ideals of the 2^m-torsion are cofinal with the powers of the coordinate ideal; thus isomorphisms of formal groups are compatible systems of isomorphisms $\mathcal{F}_A[2^m] \cong \mu_{2^m}$. By Cartier duality these are trivializations of a height-one étale 2-divisible group, and at level 2^m they form a finite étale $(\mathbb{Z}/2^m)^\times$-torsor. All the constructions used commute with base change. □

Lemma 6.4. *Let A be a source or target term of (4.1), or a finite étale algebra over such a term. For every integer m the unit induces an equivalence*

$$\tau_{\le m}(A/2) \xrightarrow{\sim} \Big(\tau_{\le m}\big(L_{K(1)}(KU_2^\wedge \wedge A)/2\big)\Big)^{h\mathbb{Z}_2^\times}, \tag{6.5}$$

natural in A, in m and in the cosimplicial direction.

Proof. By Proposition 4.4, Lemma 6.3 applies. For Adams descent, take the homotopy coefficient groups with their discrete smooth structures in the finite-continuous-set topos of [5, Lemma 3.3]; the base coefficient groups are fixed. We write $\mathrm{C}^s_{\mathrm{lc}}$ for the resulting locally constant Adams cochains. A cochain has finite image, since its parameter space $(\mathbb{Z}_2^\times)^s$ is compact and its coefficient group is discrete. That finite image lies in a common finite Igusa coefficient level, and the cochain factors through a finite quotient of $\mathbb{Z}_2^\times$. Increasing the two levels together, both occur at level 2^r. At this level the semilinear group-cochain complex is the Amitsur complex of the finite faithfully flat $(\mathbb{Z}/2^r)^\times$-torsor. Thus the locally constant augmented complex is the filtered colimit of these exact augmented complexes. This argument applies in each homotopy degree and is natural in A and in the cosimplicial maps. It gives

$$H^s_{\mathrm{cts}}(\mathbb{Z}_2^\times, \pi_j(L_{K(1)}(KU_2^\wedge \wedge A)/2)) = \begin{cases} \pi_j(A/2), & s = 0, \\ 0, & s > 0, \end{cases}$$

and similarly after tensoring with a finite projective module. For a spectrum bounded above by m, the homotopy fixed point spectral sequence in total degree d has only the finitely many terms with $0 \le s \le m - d$, so it converges, and it collapses to the edge. This gives (6.5). □

The mod-2 version of (6.2) gives natural equivalences $\tau_{\le m}(L_{K(1)}\big((L_{K(3)}(E_3^{\wedge(q+1)}))^{tC_2}\big)/2) \simeq (\tau_{\le m}(T^q/2))^{hP}$ for every m. Taking inverse limits over m in either comparison recovers the untruncated descent equivalence, by left completeness and commutation of homotopy fixed points with limits.

7 The degree-two invariant of the transfer

7.1 The universal case

Let C_2 act trivially on $KU_2^\wedge/2$, let $x \in H^1(C_2, \mathbb{F}_2)$ be the nonzero class, and let $\mathbb{Z}_2^\times$ act by Adams operations. The reduced transfer of the free C_2-set of two points gives an element

$$\mathrm{tr} \in \pi_0\big((KU_2^\wedge/2)^{h(C_2\times\mathbb{Z}_2^\times)}\big),$$

of rank zero. Since $\pi_1(KU_2^\wedge/2) = 0$, it has an invariant $\mathrm{pr}_2(\mathrm{tr}) \in H^2_{\mathrm{cts}}(C_2 \times \mathbb{Z}_2^\times, \mathbb{F}_2) \cdot \beta_{KU}$, where $\beta_{KU} \in \pi_2(KU_2^\wedge/2)$ is the Bott class; $\mathbb{Z}_2^\times$ acts trivially on β_{KU} modulo 2.

Proposition 7.1. *There is a continuous character $\chi_{\mathrm{Ad}}\colon \mathbb{Z}_2^\times \to \mathbb{F}_2$ with $\chi_{\mathrm{Ad}}(-1) = 1$ such that*

$$\mathrm{pr}_2(\mathrm{tr}) = \beta_{KU}\,(x^2 + x\chi_{\mathrm{Ad}}). \tag{7.1}$$

Proof. By the Künneth formula,

$$H^2_{\mathrm{cts}}(C_2 \times \mathbb{Z}_2^\times, \mathbb{F}_2) = \mathbb{F}_2 x^2 \oplus x\, H^1_{\mathrm{cts}}(\mathbb{Z}_2^\times, \mathbb{F}_2) \oplus H^2_{\mathrm{cts}}(\mathbb{Z}_2^\times, \mathbb{F}_2).$$

We determine the three components by restriction. Restricting to C_2 forgets the Adams action. There the transfer of the free C_2-set is the regular representation $1 + L$, with L the sign line, so tr is the class of $L - 1$; its invariant is the mod-2 first Chern class of L times β_{KU}, which is $\beta_{KU}x^2$. Restricting to $\mathbb{Z}_2^\times$, the C_2-set becomes two points and $\mathrm{tr} = 0$, so the last component vanishes. Hence $\mathrm{pr}_2(\mathrm{tr}) = \beta_{KU}(x^2 + x\chi_{\mathrm{Ad}})$ for some character χ_{Ad}.

To compute $\chi_{\mathrm{Ad}}(-1)$, restrict to the diagonal subgroup $C_2 \subset C_2 \times \{\pm 1\}$. It acts on $KU_2^\wedge$ by complex conjugation, so the homotopy fixed points are $KO_2^\wedge$, and after smashing with $\mathbb{S}/2$ we get $\pi_0(KO_2^\wedge/2) = \mathbb{F}_2$, because $\pi_0 KO_2^\wedge = \mathbb{Z}_2$ and $\pi_{-1}KO_2^\wedge = 0$. This maps isomorphically to $\pi_0(KU_2^\wedge/2)$, where tr has rank zero; so the restricted element is zero. Let e_1 generate $H^1(C_2, \mathbb{F}_2)$; both x and $\chi_{\mathrm{Ad}}|_{\{\pm1\}}$ restrict to multiples of e_1, the first to e_1 and the second to $\chi_{\mathrm{Ad}}(-1)e_1$. By naturality of pr_2,

$$0 = \beta_{KU}\,(1 + \chi_{\mathrm{Ad}}(-1))\, e_1^2,$$

so $\chi_{\mathrm{Ad}}(-1) = 1$. □

7.2 The transfer in the descent diagram

Pulling back along $x_P\colon P \to C_2$, the class (7.1) becomes $\beta_{KU}(x_P^2 + x_P\chi_{\mathrm{Ad}})$ in the cohomology of $P \times \mathbb{Z}_2^\times$. To push it into the descent diagram we need the counterparts of x_P and χ_{Ad} there. For x_P it is the cocycle x_{line} of Proposition 5.1. For χ_{Ad}, choose a finite level through which the continuous character χ_{Ad} factors. The corresponding quadratic quotient of the Igusa torsor is a $\mathbb{Z}/2$-torsor over $\mathrm{Spec}(k[[a,b]][a^{-1}])$ with a $\mathbb{G}_3$-action. Since $\mathrm{Spec}(k[[a,b]][a^{-1}])$ is affine, choose an Artin–Schreier coordinate w on this torsor. The deck involution sends w to $w + 1$, and

$$w^2 + w \in k[[a,b]][a^{-1}].$$

The stabilizer action commutes with the deck involution, so its descent cocycle is

$$x_{\mathrm{Ig}}(g) = gw - w \in k[[a,b]][a^{-1}] \qquad (g \in \mathbb{G}_3).$$

Choose $r \geq 0$ such that $w^2 + w \in a^{-2r}k[[a,b]]$. Because $g(a)/a \in k[[a,b]]^\times$ and $\mathbb{G}_3$ preserves $k[[a,b]]$, the identity

$$x_{\mathrm{Ig}}(g)^2 + x_{\mathrm{Ig}}(g) = g(w^2+w) - (w^2+w) \in a^{-2r}k[[a,b]]$$

and normality of $k[[a,b]]$ imply $x_{\mathrm{Ig}}(g) \in a^{-r}k[[a,b]]$ for every g. The integral normalization of this finite Igusa quotient is finite over the excellent ring $k[[a,b]]$ and is preserved by $\mathbb{G}_3$. Lemma 4.2 makes its action continuous. The element $a^r w$ is integral over $k[[a,b]]$, and

$$a^r x_{\mathrm{Ig}}(g) = \frac{a^r}{g(a)^r}\, g(a^r w) - a^r w.$$

Artin–Rees identifies the topology induced on $k[[a,b]]$ from that finite normalization with its adic topology. Thus x_{Ig} is continuous in the fixed lattice $a^{-r}k[[a,b]]$, and Lemma 4.3 places it in $Z^1_{\mathrm{cts}}(\mathbb{G}_3, \underline{k[[a,b]]}[a^{-1}])$. The element $[-1] \in \mathbb{G}_3$ acts on the height-one formal group as -1, hence on the Igusa torsor as the Adams operation ψ^{-1}, so

$$x_{\mathrm{Ig}}([-1]) = \chi_{\mathrm{Ad}}(-1) = 1. \tag{7.2}$$

Proposition 7.2. *The element* tr_C *of Proposition 6.1 satisfies*

$$\mathrm{pr}_2(\mathrm{tr}_C) = v_1\big[\mathrm{pr}_N(x_{\mathrm{line}} \smile x_{\mathrm{line}}) + x_N \smile x_{\mathrm{Ig}}\big]. \tag{7.3}$$

This is an identity in $H^2(\mathbb{G}_3, v_1\underline{N})(*)$.

Proof. The idea is to connect the universal transfer to tr_C by a zigzag of truncated descent diagrams. The zigzag preserves pr_2 by naturality, so the computation reduces to tracking x_P and χ_{Ad} through an explicit cochain complex.

Postnikov towers. The group P acts on T^q and $\mathbb{Z}_2^\times$ acts on $KU_2^\wedge$, and the two actions commute. For every m, the unit maps and Lemma 6.4, together with its analogue for (6.1), give

$$\begin{aligned}
&\big(\tau_{\leq m}(KU_2^\wedge/2)\big)^{h(P\times\mathbb{Z}_2^\times)} \\
&\quad\longrightarrow \mathrm{Tot}_q\Big(\tau_{\leq m}\big(L_{K(1)}(KU_2^\wedge \wedge T^q)/2\big)\Big)^{h(P\times\mathbb{Z}_2^\times)} \\
&\quad\xleftarrow{\sim} \mathrm{Tot}_q\, \tau_{\leq m}\Big(L_{K(1)}\Big(\big(L_{K(3)}(E_3^{\wedge(q+1)})\big)^{tC_2}\Big)/2\Big) \\
&\quad\longrightarrow \mathrm{Tot}_q\, \tau_{\leq m}(C^q/2).
\end{aligned} \tag{7.4}$$

The middle map is an equivalence: each of the two group resolutions has, in a fixed total degree, finitely many relevant terms after truncation, so the two bounded descent equivalences combine. All maps in (7.4) commute with the transition maps in m, and hence define a zigzag of Postnikov towers. The square

$$\begin{array}{ccc}
\mathbb{S}^{hP} & \longrightarrow & (T^q/2)^{hP} \\
\downarrow & & \downarrow \\
(KU_2^\wedge/2)^{h(P\times\mathbb{Z}_2^\times)} & \longrightarrow & \big(L_{K(1)}(KU_2^\wedge \wedge T^q)/2\big)^{h(P\times\mathbb{Z}_2^\times)}
\end{array}$$

of unit maps commutes, so (7.4) carries the pullback of tr to tr_C. In particular take the rows $m = 2$ and $m = 1$. Their termwise fibers are $\Sigma^2 H\pi_2$, and taking homotopy fixed points and totalizations preserves those fibers. Hence (7.4) induces the same zigzag on the second layers. Each row for $m = 1$ is coconnective, since the underlying coefficients have $\pi_1 = 0$. The lift of a rank-zero element to the fiber over the $m = 1$ row is therefore unique on π_0, as in Lemma 6.2. The commuting unit square thus identifies the second-layer images of the universal transfer and tr_C. It remains to compute the induced map on the π_2 coefficient complexes.

Torsor cocycles. We give the coefficient model for the second-layer maps in (7.4). For $m \geq 1$, let B_m be the normalization of $k[[a, b]]$ in the coordinate algebra of the joint level-4 frame cover and level-2^m Igusa cover over $\mathrm{Spec}(k[[a, b]][a^{-1}])$. It is finite, since $k[[a, b]]$ is excellent. Moreover $B_m[a^{-1}]$ is the original finite étale algebra. Give B_m its finite-module adic topology. The commuting actions of $\mathbb{G}_3$, P and $(\mathbb{Z}/2^m)^\times$ preserve this normalization. The $\mathbb{G}_3$ action is continuous by Lemma 4.2 and sends a to a unit multiple; the deck actions fix a.

The finite-level torsors commute with base change to the degree-q source coefficients. Finite projectivity of $B_m[a^{-1}]$ over $k[[a, b]][a^{-1}]$ gives

$$\begin{aligned}&\mathrm{Map}_{\mathrm{cts}}(\mathbb{G}_3^q, k[[a, b]])[a^{-1}] \otimes_{k[[a,b]][a^{-1}]} B_m[a^{-1}]\\ &\quad\cong \mathrm{Map}_{\mathrm{cts}}(\mathbb{G}_3^q, B_m)[a^{-1}].\end{aligned}$$

Indeed, choose a finite projective idempotent presentation after inverting a and clear the finitely many denominators in its structure and comparison maps. The resulting finite lattice and B_m contain suitable a-multiples of one another. Artin–Rees and Lemma 4.3 identify their localized solid objects and continuous-function modules. Taking the colimit of the Igusa levels, compact projectivity of $\mathbb{Z}[\mathbb{G}_3^q]^\blacksquare$ gives

$$\begin{aligned}\pi_0\big(L_{K(1)}(KU_2^\wedge \wedge T^q)/2\big) &\cong \varinjlim_{m,r} \mathrm{Map}_{\mathrm{cts}}(\mathbb{G}_3^q, a^{-r} B_m)\\ &\cong \mathrm{C}^q_{\mathrm{cts}}\left(\mathbb{G}_3, \varinjlim_m \underline{B_m}[a^{-1}]\right).\end{aligned}$$

The action coface uses the natural $\mathbb{G}_3$ action on the torsors, and the other cofaces and codegeneracies act on the Morava parameters. Thus the identifications are cosimplicial.

For Adams cochains, take the underlying coefficient groups in this display to be discrete. They are smooth: each element comes from a single B_m and a single fractional lattice and is fixed by the kernel of $\mathbb{Z}_2^\times \to (\mathbb{Z}/2^m)^\times$. We therefore use the triple complex with (i, j, q)-term

$$\mathrm{C}^i\Big(P,\, \mathrm{C}^j_{\mathrm{lc}}\Big(\mathbb{Z}_2^\times,\, \pi_0\big(L_{K(1)}(KU_2^\wedge \wedge T^q)/2\big)\Big)\Big).$$

The Adams variables are locally constant; the Morava variables retain the adic continuity and uniform pole bound of the preceding display. At fixed q, any Adams cochain has finite image, which lies in one common finite Igusa level. Increasing that level and the finite group quotient together identifies the augmented Adams complex with a filtered colimit of the finite Igusa torsors' Amitsur complexes, as in Lemma 6.4. After Adams descent, the P complex is the Amitsur complex of the frame torsor over the degree-q Tate algebra. These exact augmented complexes are natural in q. Since all three cohomological degrees are nonnegative, their coaugmentations give a quasi-isomorphism from the Tate coefficient complex to the total triple complex.

Take normalized cochains in each direction and the direct-sum total complex, with differential d and iterated Alexander–Whitney cup product in the order $P, \mathbb{Z}_2^\times, \mathbb{G}_3$. The Eilenberg–Zilber

comparison with the diagonal is natural and identifies the induced products in cohomology. The coaugmentations are maps of cosimplicial coefficient algebras, and the total cup product satisfies the Leibniz rule on cochains. In total degree at most 2 only finitely many triples (i, j, q) occur. Multiplication, the pairing $\underline{N} \otimes^{\blacksquare} \underline{k[[a,b]][a^{-1}]} \to \underline{N}$, and the localized trace projection give the coefficient products and module maps used below.

Pull the Artin–Schreier coordinate z of Lemma 5.2 back to the frame algebra. Its P-coboundary is x_P, its $\mathbb{Z}_2^\times$-coboundary is zero, and its coboundary in the q-direction is x_{line}, by the definition of x_{line}. Thus $dz = x_P + x_{\mathrm{line}}$. The coordinate w above is defined over the Igusa cover of the source term in (4.1) and then base-changed to T^q (Lemma 6.3); it is therefore P-invariant, and $dw = \chi_{\mathrm{Ad}} + x_{\mathrm{Ig}}$. In characteristic 2, the Leibniz rule, with cup products in their displayed order, gives

$$x_P \smile \chi_{\mathrm{Ad}} + x_{\mathrm{line}} \smile x_{\mathrm{Ig}} = d\big(z \smile x_{\mathrm{Ig}} + x_{\mathrm{line}} \smile w + z \smile dw\big),$$
$$x_P \smile x_P + x_{\mathrm{line}} \smile x_{\mathrm{line}} = d\big(z \smile x_{\mathrm{line}} + x_{\mathrm{line}} \smile z + z \smile dz\big).$$

So the class of $x_P \smile x_P + x_P \smile \chi_{\mathrm{Ad}}$ maps to the class of $x_{\mathrm{line}} \smile x_{\mathrm{line}} + x_{\mathrm{line}} \smile x_{\mathrm{Ig}}$.

The displayed primitives belong to this mixed complex. Choose a finite Igusa level containing w; the fixed frame level already contains z. Their equations give $a^2 z, a^r w \in B_m$ for some $r \geq 0$. The actions preserve B_m and change a only by a unit, so the coordinates, their translates and every product in the primitives lie in a single fractional lattice $a^{-s}B_m$ after increasing s. The Morava cochains are continuous there by Lemma 4.2 and continuous multiplication. The Adams cochains are locally constant, since their action on B_m factors through $(\mathbb{Z}/2^m)^\times$. The finite-projective comparison above identifies these cochains with the finite-level degree-q coefficients and respects cofaces and products. Thus the displayed identities hold in the total mixed complex. The torsor coaugmentations identify their cohomology classes with the asserted scalar solid Morava classes.

Projection to the cofiber. By Proposition 4.4 the Bott class maps to v_1, and on degree-two coefficients the map $L_{K(1)}\Big(\big(L_{K(3)}(E_3^{\wedge(q+1)})\big)^{tC_2}\Big)/2 \to C^q/2$ sends $v_1 f$ to $v_1 \,\mathrm{pr}_N(f)$ for a degree-zero coefficient f, termwise in the scalar solid bar modules. Since x_{Ig} takes values in $k[[a,b]][a^{-1}]$ and pr_N is linear over this ring, $\mathrm{pr}_N(x_{\mathrm{line}} \smile x_{\mathrm{Ig}}) = x_N \smile x_{\mathrm{Ig}}$, which gives (7.3). □

8 Contraction with the central involution

Lemma 8.1. *Let G be a topological group acting continuously on a topological $\mathbb{F}_2$-module M, and let $c \in G$ be central and act trivially on M. For these topological coefficients, use classical continuous cochains. For a continuous 2-cochain y put $(I_c y)(g) = y(c,g) + y(g,c)$. Then I_c sends cocycles to cocycles and coboundaries to coboundaries, and so induces $I_c\colon H^2_{\mathrm{cts}}(G,M) \to H^1_{\mathrm{cts}}(G,M)$. For G profinite and M a solid $\mathbb{F}_2$-module with continuous G-action, the scalar solid bar cochains similarly give*

$$I_c\colon H^2(G,M)(*) \longrightarrow H^1(G,M)(*).$$

More generally, let M_1, M_2 be continuous G-modules with a continuous G-equivariant bilinear pairing $M_1 \times M_2 \to M$, or, in the solid setting, a morphism $M_1 \otimes^{\blacksquare} M_2 \to M$ of solid G-modules. Write the pairing as a product and suppose c acts trivially on M_1 and M_2. For 1-cocycles $s \in Z^1_{\mathrm{cts}}(G, M_1)$ and $s' \in Z^1_{\mathrm{cts}}(G, M_2)$, taken in the classical continuous cochain complexes in the topological case and in the scalar solid bar complexes in the solid case,

$$I_c(s \smile s') = s(c)\, s' + s\, s'(c).$$

Proof. Adding the cocycle identities for y at (c,g,h), (g,c,h) and (g,h,c), and using that c is central and acts trivially, gives $(I_c y)(gh) = (I_c y)(g) + g\,(I_c y)(h)$. For a 1-cochain r one computes $I_c(\delta r)(g) = g\,r(c) + r(c)$, the coboundary of the 0-cochain $r(c)$. With $(s \smile s')(g,h) = s(g)\,gs'(h)$,

$$I_c(s \smile s')(g) = s(c)\,s'(g) + s(g)\,g\,s'(c),$$

and the cocycle identities at $gc = cg$ give $g\,s'(c) = s'(c)$, since c acts trivially on M_2. Insertion of c preserves continuity. In the solid case it is induced by maps of profinite parameter sets, hence gives maps of solid bar terms. The same cocycle identities, evaluated on each profinite test set, prove the solid statement and therefore its scalar version. □

Proof of Theorem 1.2. Let $c = [-1] \in \mathbb{G}_3$. It acts trivially on $\underline{k[[U,S]][(US)^{-1}]}$, on $\underline{N}$ and on v_1 modulo 2, and by Proposition 5.1 and (7.2), $x_{\mathrm{line}}(c) = x_N(c) = 0$ and $x_{\mathrm{Ig}}(c) = 1$. By Lemma 8.1,

$$I_c\ \mathrm{pr}_N(x_{\mathrm{line}} \smile x_{\mathrm{line}}) = 0,$$
$$I_c(x_N \smile x_{\mathrm{Ig}}) = x_N(c)\,x_{\mathrm{Ig}} + x_N\,x_{\mathrm{Ig}}(c) = x_N.$$

With (7.3) this gives

$$I_c\ \mathrm{pr}_2(\mathrm{tr}_C) = v_1[x_N] \neq 0, \tag{8.1}$$

since multiplication by v_1 is an isomorphism $\underline{N} \to v_1\underline{N}$ and $[x_N] \neq 0$ by Proposition 5.1. Hence $\mathrm{pr}_2(\mathrm{tr}_C) \neq 0$. By Proposition 4.4, $\pi_1(C^q/2) = 0$, and by Proposition 6.1, tr_C vanishes on π_0 coefficients, so Lemma 6.2 shows $\mathrm{tr}_C \neq 0$. By (4.2), $\pi_0(\mathrm{cofib}(L_{K(1)}\psi_3) \wedge \mathbb{S}/2) \neq 0$.

In particular $L_{K(1)}\psi_3$ is not an equivalence, so neither is ψ_3. Since $\mathbb{S}/2$ is finite, the cofiber of $L_{K(1)}\psi_{3,\mathbb{S}/2}$ is $\mathrm{cofib}(L_{K(1)}\psi_3) \wedge \mathbb{S}/2$, which is nonzero; so $\psi_{3,\mathbb{S}/2}$ is not an equivalence either. □

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